Female Urology

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 101	Surgical Anatomy of the Female Pelvis
Ch 108	UI & Prolapse: Epidemiology & Pathophysiology
Ch 105	Evaluation of Women With UI & POP
Ch 114	Conservative Management of UI
Ch 116	Retropubic Suspension (Burch)
Ch 118	Slings: Autologous, Biologic, Synthetic, Mid-Urethral
Ch 117	Reconstructive Surgery for Pelvic Organ Prolapse
Ch 119	Mesh Complications & Their Repair
Ch 121	Urinary Tract Fistulae
Ch 122	Bladder & Female Urethral Diverticula
Ch 77	Female Sexual Function & Dysfunction
★	Putting it together: female pelvic decisions

101 Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis

1 · Perineum and external genitalia

Why this matters: these landmarks guide urethral and fistula surgery.

- The meatus lies **1 cm** anterior to the vaginal opening.
- **Skene's (paraurethral) glands** sit beside the meatus and are the prostate homologue. Obstructed periurethral glands form a diverticulum or cyst.
- **Bartholin's gland** lies under the posterior end of each vestibular bulb (cyst or abscess).
- The **deep perineal space** (above the membrane) holds the urethral sphincter, urethrovaginal sphincter, compressor urethrae and deep transverse perineal.
- The **superficial space** (under Colles fascia) holds superficial perineal muscles, clitoris, vestibular bulbs and Bartholin's.
- The bulbospongiosus, external anal sphincter and both transverse perineal muscles converge at the **perineal body**. Episiotomy is angled laterally to spare the sphincter.
- The **labia majora have a dual supply**: superficial external pudendal (femoral) from above and posterior labial (internal pudendal) from below. So the **Martius flap** is reliable.

2 · Bony pelvis and ligaments

Why this matters: every repair anchors here, beside something injurable.

- The **ischial spine** (about level S5) is felt along the sacrum and SSL; it marks apical sutures.

Fixation point	Used for	Adjacent danger
Sacrospinous ligament (SSL): ischial spine to lower sacrum/coccyx	SSL fixation (apical prolapse)	Pudendal nerve/vessels near the spine; sacral plexus just posterior
Cooper's ligament (medial superior pubic ramus)	Burch colposuspension for stress urinary incontinence (SUI)	Obturator vessels; aberrant obturator
Obturator foramen; canal superolateral	Transobturator tape (TOT) trocar	Obturator nerve/vessels, causing groin pain or bleeding
Anterior longitudinal ligament (ALL) over S1	Sacrocolpopexy mesh	Presacral veins, middle sacral vessels, left common iliac vein, right ureter
Uterosacral ligament	Uterosacral (apical) suspension	Ureter (lateral, closest at the cervix); sacral nerve entrapment

- The pudendal nerve leaves through the **greater** sciatic foramen and re-enters through the **lesser sciatic foramen** to reach Alcock's canal.

3 · Pelvic floor muscles

Why this matters: when the levator fails, prolapse follows.

- The pelvic diaphragm is the **levator ani** and coccygeus. The levator is supplied directly by **S3–S5 (mainly S4)**.

- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *levator table: puborectalis, pubococcygeus, iliococcygeus; coccygeus NOT levator, lies on SSL*

Not in the paper answer:

- Standing, the anterior superior iliac spines and symphysis are vertically aligned, so abdominal pressure falls on bone.
- The interspinous line divides the anterior from the posterior compartment.
- Resting levator tone gives passive urinary and faecal continence.
- The **levator hiatus** transmits the urethra, vagina and rectum. The **levator plate** is a horizontal shelf for the viscera.
- Muscle or nerve injury leaves the hiatus lax, so the plate tilts and **prolapse** follows.

4 · Endopelvic fascia and visceral ligaments

Why this matters: fascial defects explain the type of prolapse.

- The **intermediate** fascial stratum (parametrium, paracolpium, pubocervical fascia) **gives most organ support**.
- The **pubourethral ligaments** run from the pubis to the bladder neck and urethra (puboprostatic homologue). With the anterior vaginal wall they form a mid-urethral hammock; weakness causes SUI.
- A vesicopelvic fascia defect at the arcus tendineus gives a paravaginal cystocele.
- The **cardinal (Mackenrodt's) ligament** carries internal iliac branches and the pelvic plexus, so dividing it at hysterectomy can cause bladder dysfunction.
- Distinguish the **infundibulopelvic ligament** (ovarian vessels, with the ureter beneath) with the **ovarian ligament** (ovary to uterus) MCQ.

DeLancey levels of vaginal support †

- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *I suspension (uterosacral–cardinal; uterine/vault prolapse, enterocele), II attachment (ATFP; anterior/posterior wall), III fusion (urethral hypermobility, distal rectocele, perineal descent)*

> **KEY POINT**
> Support ligaments are connective tissue condensed around vessels and nerves, so the best anchors (uterosacral, cardinal) are exactly where sutures injure the ureter, pelvic plexus and sacral nerves. *See Ch 108*

5 · Urethra and continence anatomy

Why this matters: closure anatomy explains slings and oestrogen.

- The urethra is **4.7–5.1 cm** long, behind the symphysis, hung from posterior pubourethral ligaments, on the anterior vaginal wall.
- The submucosa has a **spongy vascular plexus**; the epithelium turns from transitional to squamous distally.
- **The mucosa and submucosa generate most closure pressure and are oestrogen-dependent**, which is the rationale for vaginal oestrogen.
- The **internal sphincter** is smooth (continuing the detrusor), autonomic, at the bladder neck and proximal urethra.
- The **external sphincter** is striated, horseshoe-shaped at the mid-urethra, voluntary (pudendal), over the distal two-thirds.

- The striated complex comprises sphincter urethrae, **compressor urethrae** (closes the urethra against the anterior vaginal wall) and **urethrovaginal sphincter** (narrows the hiatus).
- **Hypermobility** is one cause of SUI. An autologous pubovaginal sling sits at the **bladder neck**; a synthetic mid-urethral sling (MUS) sits at the **mid-urethra**.

6 · Bladder and ureter

Why this matters: the ureter is the structure most often injured in pelvic surgery.

- The **dome** is peritonealised, so intraperitoneal injury needs formal repair. Injury to the **base** (on anterior vagina and lower uterus) causes vesicovaginal or vesicouterine fistula.
- The **pelvic ureter** crosses the iliac bifurcation, passes behind the ovary to the broad ligament base, runs **under the uterine artery** ("water under the bridge"), then crosses the anterior fornix to the bladder.

ALGORITHM

Ureteric danger points:

1. Pelvic brim, at the infundibulopelvic ligament.
2. Under the uterine artery (hysterectomy).
3. Beside the cardinal/uterosacral ligaments near the cervix.
4. Anterior vaginal fornix (vaginal surgery).
5. Sacral promontory: the right ureter (sacrocolpopexy).

7 · Nerves at surgical risk

Why this matters: most such nerve injury is avoidable.

- The **pudendal nerve (S2–S4)** crosses the SSL/coccygeus **close to the ischial spine** and enters Alcock's canal. Branches: inferior rectal, perineal, dorsal nerve of clitoris.
- **Pudendal block** is given transvaginally at the SSL just behind the spine.
- In SSL fixation, place sutures about 2 cm medial to the spine, through (not around) the ligament †.

Nerve	Procedure / mechanism	Deficit
Common peroneal	Lithotomy: compression at the fibular head	Foot drop
Sacral plexus (L4–S4)	SSL fixation (plexus exits just posterior to the SSL)	Posterior leg motor/sensory loss
Pudendal	SSL fixation near the spine	Perineal pain, sensory loss, sphincter dysfunction
Sacral trunks S1–S4	Uterosacral suspension (deep sutures)	Buttock/posterior thigh pain
Ilioinguinal (L1), genitofemoral (L1–L2)	Hernia repair, caesarean, hysterectomy	Mons/labial numbness or pain
Obturator (L2–L4) †	TOT trocar, obturator lymphadenectomy	Groin pain, adductor weakness
Posterior femoral cutaneous (S1–S3)	Gluteal pressure, pelvic tumour	Perineal sensory loss

KEY POINT

Post-lithotomy foot drop is peripheral (fibular head), not pelvic. Leg pain and swelling after long lithotomy suggest **compartment syndrome**, an emergency († fasciotomy).

8 · Vessels and lymphatics

Why this matters: pelvic bleeding can be torrential.

Internal iliac	Anterior division	Posterior division
Visceral	Obliterated umbilical, superior vesical, uterine, vaginal, middle rectal	None
Parietal	Obturator, internal pudendal, inferior gluteal	Iliolumbar, lateral sacral, superior gluteal

- The obturator artery may arise from the inferior epigastric.
- The **internal iliac** has rich collaterals, so ligation or embolisation is tolerated; the **external iliac** does not, so ligation threatens the limb.
- The right ovarian vein drains to the inferior vena cava, the left to the left renal vein.
- The **presacral plexus** is valveless and under high pressure (worse in lithotomy). A 1–2 mm vein can bleed more than **1 L/min**.

ALGORITHM

Sacrocolpopexy

1. Expose the ALL at the promontory over the L5/S1 disc.
2. Open the peritoneum distally.
3. Suture into the ALL over **S1** (strongest; less discitis).
4. Avoid the left common iliac vein, middle sacral vessels, presacral plexus and right ureter (each about 2.7 cm from the mid-promontory).

Organ	First nodes
Distal ⅓ urethra, lower ⅓ vagina, vulva	Inguinal, then pelvic
Proximal ⅔ urethra, upper ⅔ vagina	Iliac, obturator (pelvic)
Bladder, uterus	Perivisceral, then pelvic, presacral and great-vessel nodes
Ovary MCQ	Along the gonadal vessels to para-aortic nodes (bypasses the pelvis)

9 · Endoscopic anatomy

Why this matters: a fixed order avoids missing a diverticulum or mesh.

ALGORITHM

1. Distal urethra: glands, diverticular ostia, mesh, stones.
2. Mid-urethral coaptation (external sphincter).
3. Bladder neck coaptation.
4. Trigone and orifices (30° lens).
5. Survey the whole mucosa.

- Reduce marked prolapse with a vaginal sponge stick.

- Squamous metaplasia and cystitis cystica are benign; biopsy if in doubt.

10 · Imaging

Why this matters: imaging shows what examination misses.

- **Translabial ultrasound** shows bladder-neck mobility and prolapse, and **mesh, MUS and bulking agents**; 3D adds an axial view.
- On **endoanal ultrasound**, an internal sphincter defect is a bright break in dark muscle; an external sphincter defect is a dark lesion in bright muscle.
- **Magnetic resonance imaging (MRI)** is the fullest radiation-free test. **T2-weighted MRI is best for urethral diverticulum** and separates other cysts (Gartner, Skene, Bartholin, müllerian, epidermal inclusion).
- **MR defecography** grades descent below the **pubococcygeal line** (inferior symphysis to last coccygeal joint). Use it for defaecatory dysfunction or symptoms out of proportion to examination.

108 Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

1 · Definitions and terminology

Why this matters: the type of UI decides treatment.

- **Urinary incontinence (UI)** is complaint of involuntary urine loss (International Continence Society/International Urogynecological Association, ICS/IUGA, 2010).
- Lower urinary tract symptoms (LUTS) are **non-specific**, so combine **signs**, **symptoms** and **observations** (diary, urodynamics [UDS]).
- **Bother drives treatment**: without quality-of-life (QoL) impact, none may be needed.

Type	Complaint of loss…	Urodynamic correlate / trap
Stress UI (SUI)	On effort, cough, sneeze	Urodynamic SUI: leak with abdominal pressure rise **and no detrusor contraction**
Urgency UI (UUI)	With urgency	**Detrusor overactivity (DO) not required**; DO incontinence (DOI) is a separate UDS diagnosis
Mixed UI (MUI)	With urgency and effort	Urge- or stress-predominant, or equal
Overflow UI	Frequent small-volume loss with retention	Underactivity or obstruction; high post-void residual (PVR)
Functional UI †	Impaired mobility/cognition, normal lower urinary tract	Frail elderly, long-term care
Continuous UI	Day and night, few voids	Suspect fistula
Nocturnal enuresis	During sleep	Differs from nocturnal UUI (wakes, cannot reach toilet)

- **Urgency** is a sudden compelling desire, hard to defer (pathological); **urge** is normal desire. In **increased sensation** desire is early but voiding can be postponed.
- **Overactive bladder (OAB)** is urgency with or without UUI, usually with frequency and nocturia; it is **defined by urgency**. Frequency ≥ **8 voids/24 h** is abnormal once polyuria is excluded.

2 · Epidemiology of female UI

Why this matters: UI often starts in pregnancy, a window for prevention.

- Incidence at 40–60 years is 1–10%/year; remission up to about 12%/year, **lowest for MUI**.
- **Pregnancy** is the commonest cause of new UI in young women (about **40%**, mostly SUI). Over 30% leak at 1 year; UI at 3 months usually persists. Pelvic floor muscle training (PFMT) during and after pregnancy reduces UI.
- **Hysterectomy** can cause or cure UI; vaginal and laparoscopic routes carry more risk than open.

3 · Risk factors for female UI

Why this matters: some risks are modifiable, and each suggests a UI type.

Factor	Key point
Ageing	Strongest independent non-modifiable factor. SUI peaks at 40–59 years; UUI/MUI dominate after 60. Long-term care: over 40% at admission
Vaginal delivery	Causes later **SUI, not UUI**. Caesarean section (CS) protects only if **every** delivery is by CS; number needed to treat (NNT) is **13** to prevent one moderate–severe UI. Elective CS protects more than CS in labour. Baby ≥4 kg: odds ratio (OR) about 1.5.
Parity	First delivery OR about 1.5, then modest rise; strongest for SUI
Obesity	Body mass index (BMI) ≥30 more than doubles risk (SUI, MUI). Dose/duration-dependent and **reversible** with weight loss. Little effect on SUI surgery success
Menopause / oestrogen	SUI rises across the transition. **Systemic oestrogen increases UI** (new UI nearly doubled vs placebo). Vaginal oestrogen is not linked to SUI and treats atrophy and recurrent urinary tract infection (UTI)
Race	SUI commoner in White than Black/Asian women (Black women have higher closure pressure)
Smoking	Urgency, frequency, UUI (dose-dependent)
Caffeine	UUI, MUI, OAB, but not SUI
Diabetes	Up to 2×; neuropathy and macroalbuminuria are independent factors

KEY POINT

Obstetric injury and loss of support cause **stress** UI; ageing, neurological, metabolic and lifestyle factors cause **urgency**. So the dominant type moves from stress to urge with age.

Male UI (contrast)

- In men **UUI is commoner than MUI, which is commoner than SUI**. Male SUI is nearly always iatrogenic (radical prostatectomy [RP], pelvic surgery) or neurological. Remission reaches **40%**.
- Post-RP UI is 8–60% by definition, less with bladder-neck, nerve and Retzius sparing and in younger men. After transurethral resection of the prostate (TURP) it is about 1%.

4 · Physiology of continence

Why this matters: UI is a failure of storage or of the outlet.

Low-pressure storage

- During filling, **pelvic** parasympathetic outflow is suppressed. **Hypogastric** sympathetic β-action relaxes the detrusor and α-action closes the bladder neck. The **pudendal** nerve (Onuf's nucleus) gives tonic sphincter closure; suprapontine centres inhibit the pontine reflex until voiding is appropriate.
- **Poor electrical coupling** of detrusor cells stops local contractions spreading. The extracellular matrix (ECM) maintains compliance (altered by inflammation, radiation).

Outlet

- Outlet parts: **mucosa** (seal), **internal sphincter** and **distal rhabdosphincter**.
- In **women**, the continence zone is the **proximal and mid-urethra**; the bladder neck is often incompetent in continent women. The oestrogen-sensitive spongy submucosa gives up to **30%** of closure.
- **Support**: the anterior vaginal wall is the firm backing, and pubourethral ligaments limit proximal movement. Lax vaginal support with intact pubourethral ligaments causes **shear** in

the continence zone.

- **Male vs female**: the male bladder neck is powerful (α-tone, prostate) and the outlet fixed, so losing one unit is tolerated (TURP). The female bladder neck is weak and the outlet mobile, so external sphincter deficiency causes UI.

> **KEY POINT**
> The female outlet relies on a **single** mid-urethral sphincter against a mobile backing, so continence needs support, intrinsic urethral properties and muscle together.

5 · General pathophysiology: bladder vs outlet

Why this matters: the bladder-versus-outlet split guides work-up.

Mechanism	Causes	Result
Loss of suprapontine inhibition	Stroke, multiple sclerosis (MS), Parkinson; early diabetes mellitus (DM)	Neurogenic DO, causing UUI
Outlet obstruction	Women: after anti-incontinence surgery; men: bladder outlet obstruction (BOO)	DO with new (de novo) UUI
Detrusor underactivity	Conus/cauda lesion, neuropathy (DM, alcohol), radical hysterectomy, abdominoperineal resection (APR), opioids including tramadol	Retention, then **overflow UI**
Poor compliance	Pelvic radiotherapy	High detrusor leak point pressure (LPP), causing UI
Sphincter deficiency	RP, urethral/anti-incontinence surgery, cord injury, α-blockers, muscle relaxants	Intrinsic sphincter deficiency (ISD), causing SUI

- In **DM**, early polyuria and UUI give way to lost sensation and contractility, causing poor emptying, UTI and overflow.
- Multiple system atrophy destroys the intrinsic sphincter, so ≥20% leak after TURP; counsel.
- In post-RP UI, **ISD predominates (>70%)**; DO or poor compliance contributes in >30%.
- Each failed urethral or anti-incontinence operation adds fibrosis and ISD.
- **Labour** causes fascial injury, ischaemia and **pudendal denervation**.

6 · Pathophysiology of female SUI

Why this matters: each theory produced an operation.

- P3 P1 Female pelvic floor, continence mechanism, classification and management of SUI — *theories table: pressure transmission (Enhorning), hammock (DeLancey; repositioning not essential), integral theory (Petros–Ulmsten, giving MUS), ISD (McGuire)*

Not in the paper answer:

- In integral theory, the pubococcygeus pulls the vagina forward against the pubourethral ligaments; collagen/elastin laxity causes SUI.
- Support does not explain everything: some women with marked hypermobility are dry, and some with a fixed (scarred or neurogenic) urethra leak severely.
- Subtler ISD comes from birth ischaemia and pudendal damage.
- In the urethral hanging theory, the proximal urethra hangs below a less mobile bladder neck and pulls it open (funnelling). Integral theory also attributes urgency to premature stretch-receptor

firing.

Hypermobility vs ISD

- ISD was described with the autologous pubovaginal sling, usually after previous surgery; it equals **type III SUI** (Blaivas) MCQ.
- A **"pipestem" urethra** is fixed, non-coapting and usually iatrogenic. Subtler ISD coexists with hypermobility in most SUI.
- **Valsalva leak point pressure (VLPP)** is the abdominal pressure that causes leakage (invalid during detrusor contraction or poor compliance). **VLPP < 60 cm H_2O means ISD**, historically treated by bulking or pubovaginal sling.

> **KEY POINT**
> **Slings work in hypermobility but Burch fails in ISD**. Almost all SUI includes some ISD, so sphincter-directed procedures (pubovaginal sling [PVS], mid-urethral sling [MUS], bulking) work in hypermobile women. Burch corrects only hypermobility, so it is weak with severe ISD and a fixed urethra.

7 · OAB pathophysiology (brief)

Why this matters: post-operative urgency often has a fixable outlet cause.

- Neurogenic causes (§5) and outlet obstruction (BOO, post-sling) both lead to DO and de novo urgency.
- † Non-neurogenic theories are **myogenic** (altered excitability and coupling), **urotheliogenic** (mediator release, afferent sensitisation) and **afferent** (increased signalling).
- With pelvic organ prolapse (POP), frequency tracks anterior/apical POP and UUI tracks anterior POP severity. The main mechanism is obstruction from kinking. **OAB improves after POP repair in 60–80%**, more after anterior/apical repair; no reliable predictor.

8 · Insensible incontinence

Why this matters: rarer causes are curable.

- It is usually ISD or DOI; if treatment fails, seek rarer causes (UDS may miss them).
- Post-void dribble suggests diverticulum: pelvic magnetic resonance imaging (MRI). Continuous loss with normal voids suggests ectopic ureter: upper-tract imaging. Frequent small losses suggest overflow: UDS and PVR.

9 · Pelvic organ prolapse: definition and staging

Why this matters: staging guides surgery and comparison.

- **POP** is descent of the uterus/vault and/or vaginal walls from loss of support. **Name the vaginal segment** (anterior, apical, posterior), not the presumed organ. An enterocele is a peritoneal sac with bowel in the rectovaginal space.
- The most specific symptom is a **visible or palpable bulge**; it correlates best once past the hymen.
- Never attribute postmenopausal bleeding or pain to POP. A **pessary trial** is diagnostic: relief predicts benefit from surgery.

POP-Q stage	Most distal point on maximal strain
0	Aa, Ba, Ap, Bp = −3; C/D ≤ −(tvl − 2)

POP-Q stage	Most distal point on maximal strain
1	> 1 cm above hymen (< −1)
2	Within 1 cm of hymen (−1 to +1)
3	> +1 but < +(tvl − 2)
4	Complete eversion: ≥ +(tvl − 2)

- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *POP-Q points: hymen 0; Aa/Ap 3 cm (−3 to +3), Ba/Bp, C, D, gh, pb, tvl; 3 × 3 grid*

Not in the paper answer:

- Constipation without a bulge rarely justifies posterior repair, unless trapped stool needs splinting.
- Ask sexually inactive women if they wish to resume; it affects the repair.
- Imaging is rarely needed; dynamic MRI/defecography suits defaecatory dysfunction, enterocele or symptom–examination mismatch.
- Point Aa lies at about the urethrovesical junction. Traps: the stage is set by the single most distal point; Ba/Bp equal C in total eversion; measuring right after pessary removal underestimates.
- Baden–Walker grades 0–4 have poor interobserver reliability. The Pelvic Organ Prolapse Quantification (POP-Q) is the IUGA/ICS-recommended system.

10 · POP epidemiology and risk factors

Why this matters: treat symptoms, not findings.

- Loss of support is seen in 43–90%; **asymptomatic stage 1–2 is normal**. Symptomatic POP affects 3–12%.
- **Anterior POP is commoner than posterior, which is commoner than apical**. Apical descent is under-diagnosed: it accompanies most significant anterior POP and is universal at stage 4. **So when planning cystocele repair, assess the apex and consider vault suspension.**
- POP progresses slowly and often regresses early. Progression predictors are age >60, Ba ≥4 cm and genital hiatus (gh) ≥5 cm.

Risk factor	POP	UI (contrast)
Vaginal childbirth	Strongest; rises steeply with parity; forceps worse. Elective CS risk is about nulliparous; CS in labour about vaginal	SUI, not UUI
Age	About 40% rise per decade; surgery peaks in the 7th decade	Shift to UUI/MUI
Obesity	About 1.4–2.5-fold	>2-fold, reversible
Hysterectomy	Raises risk, most if done for POP (no vault suspension)	Cause or cure
Race	Hispanic highest, Black lowest	White highest
Oestrogen / menopause	Not consistently associated	Oral oestrogen increases UI
Other	Family history (about 5× in siblings), connective-tissue disease, chronic constipation (2.5×)	—

- Recurrence after surgery is predicted by preoperative stage 3–4.

11 · Pathophysiology of POP

Why this matters: muscle fails first and ligaments later, explaining recurrence.

- The **levator ani** is the primary support; the uterosacral–cardinal ligaments hold the organs in position over it.
- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *DeLancey I suspension (uterine/vault, enterocele), II attachment to ATFP (lateral or central anterior defect), III fusion (urethral hypermobility, distal rectocele)*
- The arcus tendineus fasciae pelvis (ATFP) runs from the pubis (1 cm from the midline) to just above the ischial spine. The commonest cystocele defect at surgery is ATFP detachment from the ischial spine.
- Defects are rarely isolated; level I loss correlates strongly with anterior POP, so the apex is the keystone.

ALGORITHM

POP sequence

1. Labour causes levator trauma or denervation.
2. The muscle loses tone and the urogenital hiatus widens.
3. Fascia and ligaments now carry the whole load.
4. They stretch and fail (ageing, straining, hysterectomy): POP.
5. POP widens the hiatus further.

- MRI shows levator defects in up to **20%** of primiparas and none in nulliparas. **Avulsion** predicts advanced, multicompartment POP.
- Electromyography shows levator neuropathy in about a quarter after vaginal delivery or CS in labour, almost none after elective CS. So **labour itself is the injury**.
- Collagen type I:III ratio is altered; Ehlers–Danlos and Marfan increase risk.

12 · POP and UI interaction

Why this matters: prolapse can hide SUI until repair.

- 40–55% of women with SUI have significant anterior POP, as do about 20% with OAB.
- **Occult SUI** is SUI seen only with the prolapse reduced (up to a quarter of advanced POP).

ALGORITHM

1. A cystocele kinks the urethra, and the obstruction masks SUI.
2. POP repair relieves the kink, so SUI is unmasked.
3. Detect it by UDS or stress test **with the prolapse reduced** (pessary or speculum).

- De novo SUI after POP surgery is predicted by age >66, DM and low urethral closure pressure.
- POP missed at sling surgery means later prolapse surgery, so examine for POP in every UI evaluation.

13 · Landmark trials

Trial	Design	Result	Take-home
CARE	Stress-continent women, abdominal sacrocolpopexy with or without Burch	Burch reduced postoperative SUI; SUI on UDS with prolapse reduced predicted it	Prophylactic Burch at sacrocolpopexy prevents de novo SUI

Trial	Design	Result	Take-home
OPUS	Stress-continent women, vaginal POP repair with prophylactic MUS vs sham	UI at 1 year 27% vs 43% (NNT about 6); more bladder perforation, UTI, bleeding, retention	Fewer de novo SUI, more adverse events, so counsel

105 Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

1 · Definitions and risk factors

Why this matters: anatomical prolapse is common; disease means symptoms.

- Risk factors for **urinary incontinence (UI)** are age, obesity, smoking, comorbidity and postmenopausal hormones; vaginal delivery causes stress UI (SUI).
- **Pelvic organ prolapse (POP)** is commonest anteriorly, then posteriorly, then apically. Half of parous women have anatomical POP; disease means **symptoms attributable to it**. A **vaginal bulge** is the most specific symptom and most likely to resolve after repair.

Category	POP risk factors
Primary only	Older age, high body mass index (BMI), high birth weight, levator defects, low education, poverty
Recurrence only	**Younger age**, preoperative Pelvic Organ Prolapse Quantification (POP-Q) stage 3–4
Both	Vaginal delivery, parity, levator hiatal ballooning
Protective (primary)	Caesarean delivery; **smoking** MCQ

- **International Continence Society (ICS) UI terms** follow the trigger: SUI, urgency UI (UUI), mixed UI (MUI). **Nocturnal enuresis** is involuntary *voiding* in sleep; **disability-associated**: cannot reach the toilet; **insensible**: unaware; **occult SUI**: only after POP reduction. Also coital, continuous, postural.

2 · History

Why this matters: history types the UI and finds reversible causes.

- Ask directly; women rarely volunteer UI. Exclude reversible causes with **DIAPPERS**: Delirium, Infection, Atrophic vaginitis, Psychological, Pharmacological, Excess urine output, Restricted mobility, Stool impaction.
- **SUI is urethral failure, UUI is bladder failure, and MUI is both**. In MUI, treat the more **bothersome** component first. A negative SUI history does not exclude SUI.
- Lower urinary tract symptoms (LUTS) may precede neurological diagnosis.
- **POP** gives bulge and pressure, worse later in the day, starting once it reaches the hymen. **Splinting** suggests high-grade POP. Anterior POP kinks the urethra, causing voiding difficulty; posterior POP obstructs defecation. POP-associated overactive bladder (OAB) may persist after repair.
- SUI coexists in about **40%** of women with POP. Many expect repair to cure leakage, so counsel.
- Past history: neurological disease, diabetes mellitus (DM), radiation, pelvic surgery, prior UI/POP surgery or mesh, obstetric trauma. **Local** oestrogen may improve UI and **systemic** oestrogen may worsen it. Family history raises POP risk and recurrence.

Occult SUI (ICS-defined) is found in about ⅓ before vaginal POP repair. **De novo SUI** appears after POP surgery without prior occult SUI; it is not ICS-defined. Both need a shared decision on concurrent versus staged surgery.

- **Drugs**: α-agonists cause retention, and **α-blockers worsen SUI in women**. Anticholinergics, calcium-channel blockers and opioids impair emptying. Cholinesterase inhibitors cause UUI. Diuretics, sodium–glucose co-transporter 2 (SGLT2) inhibitors and lithium cause polyuria; glitazones, nocturia; sedatives, confusion. **Angiotensin-converting enzyme (ACE) inhibitors cause cough, which worsens SUI**.

3 · Physical examination

Why this matters: surgery needs SUI demonstrated and prolapse staged.

- Assess frailty before surgery at ≥65 years. In **oestrogen status**, atrophic tissue is pale, flat, dry, without rugae. Note caruncle or urethral mass.

SUI testing

- The American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (AUA/SUFU) 2023 says: **objectively demonstrate SUI with a comfortably full bladder before surgery**.
- P3 SUI: evaluation in females, role of UDS, and surgical therapy in males — *cough stress test (supine, then standing), reduce prolapse for occult SUI, Q-tip >30° = hypermobility*

Not in the paper answer:

- A cough stress test (CST) at half functional capacity (largest diary void) has high sensitivity and specificity; at a comfortably full bladder specificity is lower.
- A large prolapse kinks the urethra and masks SUI, so retest with the anterior wall supported.
- The Q-tip test is uncomfortable, now mostly for teaching and trials. A vaginal swab test is equivalent, and **visual urethral mobility** (meatal rotation < or **>30°**) is routine.

POP examination

- Examine lithotomy **and standing**, straining, for the worst prolapse.
- With a **split Graves speculum**, retract the posterior wall to see the anterior; rotate to see the posterior; use both blades to isolate the apex.
- If the blade is expelled with posterior foreshortening, there is **level I loss, meaning vault prolapse**; if it stays, isolated rectocele or enterocele.

POP-Q system

- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *POP-Q points table: hymen 0; Aa/Ap 3 cm (−3 to +3), Ba/Bp, C, D (absent after hysterectomy), gh, pb, tvl*

Not in the paper answer:

- Ba/Bp range from −3 to +tvl. The genital hiatus (gh) runs from mid-meatus to posterior hymen; the perineal body (pb) from posterior gh to mid-anus. Total vaginal length (tvl) is measured with the prolapse reduced. Stage is set by the single most distal point.
- A rectocele (posterior wall into vagina) differs from rectal prolapse (mucosa out through anus).

Stage	POP-Q criteria	Baden–Walker grade (by eye)
0	Aa, Ba, Ap, Bp −3; C or D ≤ −(tvl − 2)	0: normal
I	Leading edge < −1 cm	1: upper vagina to midplane
II	Leading edge −1 to +1 cm	2: midplane to hymen
III	> +1 cm but < +(tvl − 2)	3: through/beyond hymen
IV	≥ +(tvl − 2): complete eversion	4: complete eversion

KEY POINT
Stage II is a 2-cm window around the hymen; the III/IV boundary depends on tvl. Example: tvl 8, Bp +5 lies between +1 and +6, so **stage III**.

Other elements

- The **bulbocavernosus reflex** (S2–S4) is present in only about **70%** of normal women, so its absence alone is not pathological.
- For **prior mesh**, use International Urogynecological Association (IUGA)/ICS terms: *prominence* (no breach) versus *separation* (breach); *extrusion* (through vaginal epithelium) versus *perforation* (into bladder, urethra or another viscus).

4 · Supplemental evaluation

Why this matters: the uncomplicated patient needs few tests.

Urinalysis (UA) and post-void residual (PVR) are universal. For the AUA/SUFU 2023 index patient having SUI surgery, **UA, PVR and objective SUI demonstration** are required; cystoscopy, urodynamics (UDS) and imaging are not routine. In non-neurogenic OAB only UA is required (PVR optional).

- P3 SUI: evaluation in females, role of UDS, and surgical therapy in males — *non-index features needing more tests: SUI not shown, neurogenic, urgency-predominant MUI, raised PVR, stage ≥3 POP without SUI, failed surgery; questionnaires; VALUE*

Not in the paper answer:

- Unexplained haematuria or pyuria is also non-index.
- Pad counts do not correlate with 24-hour pad weight (grading below).

Symptom quantification

- A 3-day **bladder diary** is both diagnostic and therapeutic *See Ch 114*. The largest void equals functional capacity.
- The International Consultation on Incontinence Questionnaire–UI Short Form (ICIQ-UI SF) correlates with pad tests. For POP, use the Pelvic Floor Distress Inventory and Impact Questionnaire (**PFDI / PFIQ**).
- **Pad tests** are mainly for research. ICS 1-hour test: 500 mL drunk in <15 min, standard activities, then the pad is weighed.

	1-hour test	24-hour test
Normal	<1.4 g	<4.4 g
Mild	1–10 g	4–20 g
Moderate	11–50 g	21–74 g
Severe	>50 g	≥75 g

- **Double-dye tampon test** (oral phenazopyridine and intravesical methylene blue): orange means ureterovaginal, blue vesicovaginal fistula.
- UA looks for haematuria, pyuria, glucosuria; do not treat asymptomatic bacteriuria.
- Measure **PVR** by bladder scan and repeat it (one value is unreliable). There is no agreed cut-off; asymptomatic women are almost always <100 mL. A raised PVR changes counselling before SUI surgery or antimuscarinics.
- **UDS** are not needed in the index SUI patient *See Ch 107*.

- **Cystoscopy** is not routine; use it for haematuria, urgency or irritative LUTS, recurrent urinary tract infection (UTI), prior sling or mesh, and pelvic radiation.

Imaging

Modality	Main indication
Voiding cystourethrogram	Reflux, diverticula, cystocele
Renal ultrasound	Hydronephrosis: chronic retention, **severe POP (ureteric kinking)**, fistula, neurogenic bladder
Translabial ultrasound	**Modality of choice for mesh** (computed tomography and magnetic resonance imaging [MRI] poor); levator defects
MRI / dynamic MRI	**Urethral diverticulum (standard of care)**; complex or recurrent POP; symptoms out of proportion to findings

- A diverticulum gives the **3 Ds**: dysuria, postvoid dribbling and dyspareunia *See Ch 122*.
- MRI **HMO** system: H-line = hiatal width, M-line = levator plate descent, O = organ beyond the H-line.

5 · Management pathway

Why this matters: only red flags need early specialist work-up.

ALGORITHM

1. History, examination, UA and PVR.
2. Exclude DIAPPERS causes, UTI, obstruction and tumour.
3. Classify the UI and grade the bother.
4. If there are red flags (pain, haematuria, recurrent UTI, pelvic mass, prior radiation or pelvic surgery, prolapse beyond the introitus, fistula), then refer for specialist work-up.
5. If not, start initial management and reassess.
6. Add tests only if they change management; then surgery.

Diagnosis	Initial management
SUI	Lifestyle and behavioural therapy including pelvic floor muscle training (PFMT) *See Ch 114*; † pessary; surgery if these fail
UUI / OAB	Lifestyle, behavioural therapy; antimuscarinic trial *See Ch 113*
Significant PVR	Treat constipation, review drugs; catheter drainage if PVR 200–500 mL, then reassess
Symptomatic POP	† Observation, PFMT, pessary or surgery, chosen by bother and goals

6 · Landmark trials

Trial	Design	Result	Take-home
† VALUE	Randomised controlled trial, uncomplicated demonstrable SUI: office evaluation with or without UDS	Office evaluation alone non-inferior	Preoperative UDS not needed in index SUI

114 Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

1 · Definition and principles

Why this matters: every guideline puts these first.

- **Conservative treatment** is any therapy that is neither surgery nor drugs.
- **Behavioural treatment** changes behaviour or environment, or teaches skills. It is effective and **essentially risk free**, so it is first-line in all guidelines for stress (SUI), urgency (UUI) and mixed (MUI) urinary incontinence, overactive bladder (OAB) and nocturia.
- Programmes are multicomponent and individualised: education, diary, pelvic floor muscle training (PFMT), "the knack", urge suppression, bladder training (BT), biofeedback (BF), electrical stimulation (E-stim), scheduled voiding, and fluid, caffeine and weight advice. Nurses and advanced practice providers (APPs) deliver most of it.
- Outcome is unpredictable, so offer it to anyone willing; dementia patients benefit from caregiver-led programmes.

	Outlet-directed	Bladder-directed
Target	Urethral closure	Bladder function via voiding habits
Examples	PFMT; behavioural training with urge suppression	BT; delayed voiding

	Patient-dependent	Caregiver-dependent
Examples	BT, PFMT, behavioural training	Timed voiding, habit training, prompted voiding
Needs	Intact function, learning, motivation	Knowledgeable, motivated caregiver
Suits	Cognitively and functionally intact	Impaired, nursing home

KEY POINT

Why first-line if surgery cures SUI more often? Conservative therapy is risk free and at least as effective as drugs. This asymmetry of harm drives every recommendation.

2 · Assessment

Why this matters: you look for habits you can change.

- History targets *triggers and habits* (onset, progression, triggers).
- A **bladder diary for 3–7 days** records intake, voids and volumes, urgency, leaks and pads. It estimates **functional capacity**, and its **self-monitoring effect** makes it assessment, outcome measure and treatment at once.
- The **Questionnaire-Based Voiding Diary** is a validated recall alternative, taking 5–7 min at the first visit.
- **Education** comes first: anatomy and the mechanism of the patient's own leakage. Improvement is gradual and needs daily practice.

3 · Lifestyle modifications

Why this matters: these are cheap adjuncts; evidence is weaker than for skill-based therapy, except weight loss.

Fluid management

- **Inadequate intake is more common** than excess. Self-restriction causes dehydration, urinary tract infection (UTI), constipation, concentrated irritant urine and lower capacity.
- Excess intake (~3.7 L/day) worsens frequency, so reduce it if 24-h output is >~ **2100 mL**.
- **Target** six to eight 8-oz glasses, or **30 mL/kg/day**, minimum **1500 mL**, at set times. The Institute of Medicine (IOM) figure of 2.7 L/day includes water from food.
- Change the **timing**, not the total: stop fluids **3–4 h before bed** for nocturia. For oedema-driven nocturia, use stockings or leg elevation in the late afternoon, or a loop diuretic mid-to-late afternoon.
- Cutting intake by about ¼ reduces frequency, urgency and nocturia. The goal is a **normal range, not restriction**.

Caffeine and irritants

- Caffeine is a **diuretic and irritant**. It raises filling detrusor pressure and is a risk factor for detrusor overactivity (DO).
- About **250 mg/day** (≈2 cups) is associated with moderate–severe UI in men; coffee is not linked to nocturia.
- Only a subset is caffeine-sensitive. Try a 3–5-day reduction, tapering with decaf to avoid withdrawal headache.
- Other suspects are aspartame, citrus, spicy food and carbonated or artificially sweetened drinks. Eliminate one at a time with a food diary.

Bowel function

- Constipation (<3 stools/week) and straining are linked to lower urinary tract symptoms (LUTS). The reasons are shared innervation, cross-sensitisation, and straining that causes **pudendal denervation**.
- **Faecal impaction** can cause OAB or overflow.
- Advise fibre **with adequate fluid**, exercise, and regular evacuation after meals (gastrocolic reflex).

Obesity, smoking, diet

- Obesity roughly doubles UI risk. **Loss of 5–10% body weight** sustained for 12 months substantially reduces UI, mainly SUI. Bariatric surgery resolves UI in morbid obesity.
- Behavioural weight loss in overweight women is the **only grade A lifestyle recommendation**.
- For smoking, no UI recommendation is possible (grade D); advise cessation for general health (grade C in men).
- A fruit, vegetable and whole-grain diet helps prevent UI after menopause (B). Vitamin D reduced UUI in postmenopausal Black women (B).

4 · Pelvic floor muscle training (PFMT)

Why this matters: most failures come from a wrong contraction.

- PFMT is the cornerstone for SUI, UUI, MUI, urgency and frequency (**Arnold Kegel, 1948**) MCQ.
- **Mechanism**: contraction lifts the urethra and compresses it against the symphysis. Training causes striated muscle **hypertrophy**, which raises urethral pressure and improves bladder-

neck support. There is also a **skill** route (the knack).

Assessment of PFM function

- A correct contraction gives **perineal elevation**; perineal descent means the patient is straining.
- On **digital vaginal exam**, levator ani lies just above the hymen at **4 and 8 o'clock**. Assess tone, symmetry, hold and quick flicks.
- Scales: **Modified Oxford** is the most used; **Brink** scores pressure, duration and displacement. Per anum, puborectalis lies 2.5–4 cm from the verge.
- Note co-contraction, and **overactivity** (hypertonic, cannot relax) versus hypotonicity.

Teaching

- Say: "Tighten as if stopping gas, pull up and in", without using buttocks or abdomen. For men: "move the penis up and down".
- The commonest errors are **co-contraction** (rectus, gluteals) and **Valsalva substitution**. Counting aloud prevents breath-holding.
- **Verify correct contraction before prescribing.** Failure to isolate the muscle is probably the commonest cause of poor outcome; a pamphlet or "stop the stream" does not ensure it.

Regimen

No regimen is ideal; good results come with **45–60** contraction–relaxation pairs per day.

> **ALGORITHM**
> 1. Empty the bladder.
> 2. **Lying**: 5 quick (1–2 s) and 5 sustained (5, building to 10 s) contractions.
> 3. Repeat **sitting**, then **standing** (total 30).
> 4. Rest as long as each contraction (work:rest **1:1 to 1:2**).
> 5. Do this twice daily = **60/day**.

- Use maximal effort for each contraction, and progress from reclining to upright.
- **The knack**: contract **just before and during** a cough, sneeze or lift. It reduces leakage immediately. If already leaking, squeeze anyway.

Evidence

- PFMT beats no treatment for SUI, UUI and MUI, short and long term; pure SUI does best.
- **Supervised training beats unsupervised** (grade A), and **age is no deterrent**.
- In pregnancy, antenatal PFMT in continent women prevents late-pregnancy and postpartum UI. Give supervised postpartum PFMT if UI persists ~3 months.
- **After radical prostatectomy (RP)**, PFMT hastens continence in the first 3 months. Eight weeks of behavioural training helped men leaking 1–17 years after RP. Adding E-stim or BF does not help; Pilates or yoga are not substitutes.
- For post-micturition dribble, use a strong PFM contraction after voiding, or urethral milking.
- In **neurological OAB** (multiple sclerosis, incomplete spinal cord injury, Parkinson disease, stroke), PFMT is safe and first-line (B). Transcutaneous electrical nerve stimulation (TENS) is a first-line option after stroke. PFMT adds to tolterodine for UUI.

Cones and devices

- **Weighted vaginal cones** work because slipping prompts a reflex contraction (progressive overload); use 2 × 20 min/day. They **cannot be retained with POP or a wide introitus**, and

some women retain them without contracting. They are not better than PFMT, and adding them adds nothing.
- The Colpexin sphere adds no strength. **Motion-sensing inserts** can tell lift from strain, which pressure perineometers cannot; an app-guided device beat a written or video programme at 12 months.

5 · Urge suppression

Why this matters: rushing to the toilet makes UUI worse.

- A well-timed PFM contraction **reflexly inhibits the detrusor**. Rushing raises bladder pressure, and toilet cues act as conditioned triggers.

ALGORITHM

1. When the urge comes, do not rush.
2. Stand still or sit, and relax.
3. Do several quick PFM squeezes, then distract yourself.
4. The urge subsides like a wave; then walk normally to void.
5. Once mastered, wait 5 min after the urge, then extend.

- Used alone for UUI, it gives a mean **60–80%** reduction in incontinence, and also reduces urgency, frequency and nocturia.
- For **nocturia**, suppress the urge on waking; void if it has not passed in 1–2 min.
- It is at least as good as antimuscarinics and adds to drugs. In men it lowered frequency more than an antimuscarinic with an α-blocker. In men on an α-blocker, adding behavioural therapy reduced nocturia more than adding an antimuscarinic.

6 · Biofeedback (BF)

Why this matters: BF teaches, it does not treat.

- BF is a **teaching technique**, not a treatment. It shows PFM activity in real time, by manometry or surface electromyography (EMG), and works by operant conditioning. Kegel's perineometer was the first BF device.
- EMG separates PFM from abdominal activity; pressure BF cannot separate lift from strain. On EMG, fast-twitch fibres give ~2-s spikes and slow-twitch fibres ~10-s plateaus.
- BF **adds no clear benefit to supervised PFMT**; older benefit came from extra contact. Use it for very weak PFMs, poor proprioception, an uncertain contraction, or motivation.

7 · Electrical stimulation (E-stim)

Why this matters: the settings depend on the goal, and its real role is narrow.

- Non-implanted stimulation (vaginal or anal probe, or surface electrodes) produces **PFM contraction** and **detrusor inhibition**.

	Long-term (chronic)	Maximal
Intensity	Below sensory threshold	Just below pain threshold
Aim	Detrusor inhibition via pudendal afferents	Urethral closure (direct and reflex striated contraction)
Dose	6–12 h/day for months	15–30 min, several times a week to twice daily

Indication	Frequency
UUI	**5–20 Hz** (low)
SUI (including post-prostatectomy)	**20–50 Hz** (higher)
MUI	~20 Hz or alternating
All	Pulse 200–1000 µs, rectangular, biphasic

- E-stim beats sham but **adds nothing to PFMT**, so PFMT is first-line (A). Its **niche** is women who **cannot contract voluntarily**.
- **Tibial nerve stimulation** for UUI or OAB: transcutaneous equals percutaneous in efficacy, with fewer adverse effects *(see Ch 115)*.

KEY POINT
E-stim is dosed by goal. Sub-sensory, low-frequency, prolonged stimulation inhibits the detrusor; near-painful, higher-frequency, brief stimulation contracts the sphincter. Volitional PFMT does both, so E-stim earns its place only where volitional contraction is absent.

8 · Bladder training and scheduled voiding

Why this matters: BT is grade A first-line in women; frail patients need caregiver programmes.

- P1 Describe pathophysiology of OAB. Briefly outline management & describe nonmedical non-pharmacological management — *BT breaks urgency, frequent voiding, reduced capacity cycle; void by clock not urge (dissociates voiding from urgency)*

Not in the paper answer:

- BT also works for SUI and MUI; it is grade A first-line in women, but evidence in men is insufficient.
- Its mechanism is unknown; improved cortical control and central afferent modulation are proposed.
- Adding behavioural training to drugs gives benefit that persists after the drug stops (UITN).
- BT has three components: education, an **incremental schedule**, and urge control (distraction, "I can wait", quick flicks).

ALGORITHM
1. Start with a diary.
2. Set an initial interval of ~1 h while awake (30 min if needed).
3. Void on waking, at each interval and at bedtime, but not between.
4. If leaks and urgency are controlled, increase by **15–30 min/week**.
5. Target an interval of **3–4 h**; review weekly with reinforcement.

- BT beats no treatment for UUI, SUI and MUI. The Agency for Healthcare Research and Quality (AHRQ) rates BT equal to PFMT. BT with prompting after hip fracture reduced functional UI at 6 months.
- **Delayed voiding** has no schedule: at each urge, suppress it, **wait 5 min**, and extend gradually. BT is triggered by the clock; delayed voiding by the urge plus a wait.

Caregiver-dependent toileting: elderly, nursing homes

Programme	Patient role	Key features
Timed voiding	Passive	Fixed 2–4 h by day; dryness goal. Also for impaired sensation and infrequent voiders. ~20% of frail elders become dry, 30–50% improve; a 2–3-day trial predicts success

Programme	Patient role	Key features
Habit training	Passive	Schedule individualised from the diary, shorter than the usual interval, to **pre-empt** expected leaks; suits diuretic or time-patterned UI
Prompted voiding (PV)	Responsive	**Monitoring**, then **prompting**, then **praise**; every 2–2.5 h. 25–40% of nursing-home residents respond (daytime leaks ≤1); a 3-day trial predicts
Functional incidental training	Responsive	PV plus mobility and strength exercise
Bladder training	Active	Lengthens interval; needs ability and motivation

- **Only BT lengthens the interval**; caregiver programmes aim at dryness. PV needs a consistent caregiver, and success depends on the caregiver more than the patient. Ultrasound-assisted PV beat fixed-interval PV.
- † First exclude reversible causes (DIAPPERS) and check post-void residual (PVR) *(see Ch 105)*.

9 · High-tone PFM dysfunction

Why this matters: a tight pelvic floor needs relaxing, not strengthening.

- Raised tone and poor relaxation cause pelvic or bladder pain, hesitancy, straining and incomplete emptying.
- Voiding by Valsalva provokes a reflex PFM contraction, whereas normal voiding starts with PFM relaxation.
- **Down-training**: contract, then relax, with work:rest **1:4 or longer**; BF is useful. To void: slow down, breathe, relax and wait; use double voiding.

10 · Mechanical devices

Why this matters: devices avoid surgery.

Devices are **intravaginal** (support) or **intraurethral/meatal** (occlude); no device is proven superior.

Continence pessaries

- **Rationale**: the pessary restores the proximal urethra above the pelvic floor. It **stabilises the bladder neck, lengthens the urethra and compresses it against the pubis**, like a sling. The knob of the ring sits at the urethrovesical junction.
- **Types** for SUI: incontinence **ring** (with knob), **dish** and **Hodge**. † For pelvic organ prolapse (POP), support pessaries (ring, Gehrung) and space-filling ones (Gellhorn, cube, donut) suit advanced POP or poor perineal support.
- † **Fitting**: choose the largest comfortable size, one finger-breadth from the wall. The patient walks, coughs, strains and voids before leaving; there must be no expulsion, pain or obstruction.
- Wear it continuously or only for exercise; it also helps POP-related urgency and frequency.
- **Contraindications**: active pelvic infection, severe ulceration, silicone or latex allergy, **likely non-compliance with follow-up**.
- **Complications**: discharge and odour. Neglected use causes **erosion, impaction, vesicovaginal or rectovaginal fistula**, and bleeding.

- **Care**: follow-up every **24 weeks** is acceptable (versus 12; grade C). Anaesthetic cream eases removal; oestrogen with Lactobacillus adds nothing. † Inspect for erosion at each review.
- About half improve, and a ring pessary for OAB roughly equals behavioural therapy. The International Consultation on Incontinence (ICI) says pessary or PFMT may be offered; pessary with PFMT beats PFMT alone for POP symptoms. See **ATLAS** under trials.

Other devices

- **Uresta** is a reusable bell-shaped pessary with a handle (sizes 3–5); its wide base supports the urethra. **Impressa** is a disposable, self-inserted tampon-like device giving tension-free suburethral support; it need not be removed to void.
- No urethral inserts or meatal devices are available in the USA. † An insert is a balloon plug for pure SUI (risks: UTI, haematuria, migration).
- † Use pads when treatment fails or is declined, and prevent **incontinence-associated dermatitis**. Female external catheters suit bed-bound women.
- † **An indwelling catheter is the last resort**: unmanageable retention, urine-contaminated sacral or perineal wounds, terminal care, or preference. Clean intermittent self-catheterisation (CISC) is preferred for retention.

11 · Vaginal oestrogen

Why this matters: atrophy undermines other treatments.

- For SUI with atrophy, PFMT with vaginal oestrogen beats oestrogen alone (C). † Local oestrogen improves urgency, frequency, recurrent UTI and pessary tolerance *(see Ch 113)*.

12 · Adherence, prevention and delivery

Why this matters: patients stop doing them.

- **Adherence caps efficacy** and declines over time. Remedies are realistic expectations, follow-up, reminders and linking exercise to daily cues. Predictors are self-efficacy, intention and therapist enthusiasm.
- **Prevention in older women**: group classes.
- **Prevention in childbearing**: supervised antenatal PFMT (12 weeks from 16–24 weeks' gestation) prevents late-pregnancy and early postpartum UI (A). Offer a postnatal programme after forceps or a baby ≥ **4000 g**. Antenatal PFMT does **not** prevent postnatal prolapse.
- **Prevention at prostatectomy**: PFMT before surgery, with or without early postoperative PFMT, shortens time to continence.
- **Delivery** is limited by access, not efficacy. Options are continence APPs, group classes, internet programmes (benefit maintained at 2 years) and evidence-based apps (Tät); most apps are poor. Intensity is traded for reach.

13 · Evidence grading: key ICI recommendations

Why this matters: examiners ask for grades.

Grade (Oxford)	Meaning
A	Consistent level 1 evidence; effectively mandatory
B	Consistent level 2/3 studies or majority of randomised controlled trial (RCT) evidence
C	Level 4, or Delphi-processed expert opinion

Grade (Oxford)	Meaning
D	**No recommendation possible** (not "weak")

ICI 2023 recommendation	Grade
Supervised PFMT first-line for UI in women of all ages	A
PFMT over E-stim as first-line	A
BT first-line for women with UI	A
Behavioural weight loss for overweight women with UI	A
Antenatal supervised PFMT in continent women to prevent UI	A
PFMT reduces POP symptoms (stage reduction uncertain)	A; B
PFMT over drugs first-line (fewer adverse effects, less dropout)	B
PFMT before surgery (surgery more effective but riskier)	B
Adding BF, cones or E-stim to PFMT: not routine	B/C
Fluid adequacy and caffeine reduction	B
Smoking cessation for UI	D

- AHRQ (2018): behavioural therapy is **more likely than anticholinergics to cure UUI** (high strength).

14 · Landmark trials

Trial	Design	Result	Take-home
First behavioural vs drug RCT	Older women, UUI: behavioural vs titrated drug	Behavioural fewer leaks, more satisfaction	Behavioural at least equal to drugs for UUI
MOTIVE	Men with OAB, no obstruction: behavioural vs oxybutynin extended release (ER)	Equivalent reduction in voids	Behavioural therapy works in men
UITN (†BE-DRI)	Urgency-predominant UI: tolterodine ER with or without behavioural	Combination better satisfaction, bother; persisted after drug stopped	Add behavioural training to drugs
ATLAS	Stress-predominant UI: pessary vs behavioural vs both	3 mo: behavioural better satisfaction; combination not better; no difference at 12 mo	Pessary reasonable alternative; behavioural first
Weight-loss RCT (†PRIDE)	Overweight women: 6-month intensive weight loss vs education	Greater UI reduction	Recommend 5–10% weight loss (A)
GBT	Women with UI: one 2-h bladder-health class vs none	Modestly better ICIQ-SF score to 12 months	Group classes cheap, scalable

116 Retropubic Suspension Surgery for Incontinence in Females

1 · Mechanism

Why this matters: suspension fixes only one cause of leakage.

- Stress urinary incontinence (SUI) arises from **urethral hypermobility** and **intrinsic sphincter deficiency (ISD)**, alone or combined.
- Suspension rests on **pressure transmission** (Enhorning). The hammock theory (DeLancey) adds compression against a firm backstop.

> **KEY POINT**
> Suspension corrects **hypermobility, not the sphincter**: it lifts and stabilises bladder neck and proximal urethra, so presupposes a mobile urethra. A fixed ISD urethra needs compression (sling, artificial urinary sphincter (AUS)).

2 · Patient selection

Why this matters: a mobile urethra predicts success.

- Stress test supine, then standing if negative. Q-tip >30° means hypermobility. Note pelvic organ prolapse (POP), oestrogen and incisions.
- AUA/SUFU minimum: **post-void residual (PVR) and urinalysis**; urodynamics only if leak not shown. Valsalva leak point pressure <60 or maximal urethral closure pressure <20 cm H_2O suggests ISD. For Burch, **mobility matters more than urethral function**.
- **Indications** (Burch remains an AUA/SUFU option with midurethral sling (MUS), autologous sling and bulking): SUI with hypermobility, or the stress part of mixed incontinence (MUI), not pure urgency incontinence (UUI); **concomitant abdominal surgery** (sacrocolpopexy); **difficult vaginal access**; **mesh aversion**.

Contraindications	Raise failure risk
Immobile vagina (cannot reach Cooper's)	Baseline urgency, preoperative anticholinergic
Immobile urethra (negative Q-tip ≈ **5-fold** failure): use sling or AUS	Obesity, menopause, age >60
ISD alone; "lead-pipe" (fixed, open) urethra	Prior hysterectomy or SUI surgery; advanced POP; MUI

3 · Anatomy

Why this matters: these landmarks hold every stitch.

- **Retzius** is an avascular preperitoneal space between symphysis and bladder, floored by endopelvic fascia. **Cooper's ligament** lies on the superior pubic ramus.
- The **arcus tendineus fascia pelvis (ATFP)** runs from symphysis to ischial spine; lateral vaginal detachment from it is a **paravaginal defect**.

4 · Burch colposuspension

Why this matters: a classic SAQ.

Paravaginal tissue is sutured to Cooper's; sutures elevate, **fibrosis maintains**.

ALGORITHM

1. Low lithotomy, prep including vagina, Foley.
2. Pfannenstiel or low midline incision (with or without Cherney).
3. Enter Retzius; sweep fat laterally.
4. Find bladder neck by Foley balloon and vaginal finger.
5. Dissect bladder off vagina **laterally**; expose Cooper's.
6. Place sutures, then cystoscopy.

- P1 Burch colposuspension — *2–4 sutures (0/1-0) each side, full thickness not lumen; distal at mid-urethra, proximal at bladder neck, lateral; complication rates*

Not in the paper answer:

- A missed intravesical suture later causes infection, irritative voiding, stones or fistula.
- Laparoscopic Burch may be intra- or extraperitoneal; a drain suits bleeding concern.
- Pass sutures medial to lateral and tie **distal first**, **loosely** (Tanagho), leaving **≈ 2 fingerbreadths** between bladder neck and pubis; over-tightening obstructs.

KEY POINT

Danger points: never dissect midline over urethra or bladder neck (Santorini bleeding, injury, obstruction if sutures medial). Passing through Cooper's may bleed suddenly: use pressure, clips, sutures.

Approach	Bottom line
Open (Pfannenstiel)	Cure **85–90% at 1 y, ≈70% at 5 y**
Laparoscopic (1991) / robotic	Subjective cure equivalent to open to 18 months
Needle suspension (Pereyra, Stamey, Raz)	Abandoned: sutures tore through

5 · Other retropubic procedures

Why this matters: a higher anchor means more obstruction.

Technique	Anchor	Holds by	Obstruction
Burch	Cooper's ligament	Sutures; high V-shaped angle	Greater
Marshall–Marchetti–Krantz (MMK)	Symphyseal periosteum	Sutures; high	Greater
Paravaginal (Richardson)	ATFP	Scarring; flatter	Less
Vagino-obturator shelf (VOS)	Obturator fascia (or iliopectineal)	Scarring; flatter	Less

- MMK has a unique complication: **osteitis pubis**.

KEY POINT

AUS in women is second-line for ISD (recurrent or neurogenic). Place it at the bladder neck, preferably **abdominally**; transvaginal placement raises infection and erosion.

6 · Perioperative care

Why this matters: cystoscopy catches injuries on the table.

- **Cystoscopy is mandatory**: no suture in bladder or urethra, and bilateral efflux. If efflux is absent, release sutures.

- For incomplete emptying, use clean intermittent catheterisation (CIC), preferred to suprapubic, until PVR is consistently **<150 mL**.

7 · Outcomes and complications

Why this matters: Burch is now judged against slings.

- Burch roughly equals retropubic MUS; meta-analyses slightly favour MUS, which perforates more. Burch is longer, recovers slower and causes more voiding dysfunction.
- Pubovaginal sling (PVS) is more effective but more obstructive (SISTEr). Long-term failure is ≈ 1 in 5 for both.
- With POP, add prophylactic Burch at sacrocolpopexy in stress-continent women (CARE); with existing SUI, MUS beats Burch.
- Voiding dysfunction (2–27%) is treated by CIC, then urethrolysis *(Ch 118)*. De novo overactive bladder (OAB, 8–27%) needs obstruction excluded.
- **Posterior prolapse** (2.5–27%) occurs because anterior elevation opens the cul-de-sac, so examine the posterior compartment.
- **Osteitis pubis** (MMK 0.7–2.5%) appears at 2–12 weeks with suprapubic pain to the thighs; give rest and analgesia. Osteomyelitis needs antibiotics, debridement.
- Review by 6 months (continence, voiding, OAB, sex, PVR); long-term means >48 months.

8 · Landmark trials

Trial	Design	Result	Take-home
SISTEr	Randomised trial (RCT), 655: Burch vs autologous fascial PVS	PVS higher cure at 2 and 5 y; more urinary infection, voiding dysfunction, de novo UUI	Sling more effective, more morbid
Ward–Hilton (TVT vs colposuspension)	First RCT, 348	No difference to 5 y; TVT more perforation; Burch more voiding dysfunction, later POP	MUS equals Burch, less invasive
CARE	RCT, 322 stress-continent, sacrocolpopexy with or without Burch	Less postoperative SUI with Burch, sustained 2 y	Add prophylactic Burch at sacrocolpopexy

118 Slings: Autologous, Biologic, Synthetic and Mid-Urethral

1 · How slings work — theories of continence

Why this matters: the theory decides where the sling sits and how tight it is.

- P3 Female pelvic floor, continence mechanism, classification and management of SUI — *hammock (DeLancey) vs integral theory (Petros–Ulmsten: pubourethral ligaments, suburethral hammock, pubococcygeus); PVS at bladder neck, MUS loose at mid-urethra*

Not in the paper answer:

- Supporting integral theory, incontinent women have ≈ **40% less collagen** in skin and round ligament MCQ.
- On straining, the posterior urethral wall slides away from the anterior wall (held by pubourethral ligaments), so the bladder neck funnels. In intrinsic sphincter deficiency (ISD) the urethra cannot close at rest.
- **Hammock theory (DeLancey):** in stress urinary incontinence (SUI) the suburethral layer, attached laterally to the arcus tendineus fasciae pelvis (ATFP), is deficient. The hypermobile urethra then receives pressure unequally and leaks.
- **Integral theory (Petros–Ulmsten):** maximal urethral closure pressure (MUCP) is at the mid-urethra. Injury to any component (childbirth, surgery, ageing, oestrogen loss) causes SUI.
- **Urethral hypermobility is a SIGN** of damaged support, **not the cause** of SUI.
- A pubovaginal sling (PVS) sits under **minimal tension at the bladder neck** and rebuilds the hammock. A MUS lies **loose under the mid-urethra**; it arrests the posterior wall and aids coaptation. Neither should obstruct at rest.

Historical milestones MCQ

- McGuire reintroduced the fascial PVS (1978); Blaivas used a short free rectus graft with adjustable tension.
- The intravaginal slingplasty (1995) became the **tension-free vaginal tape (TVT)**, the first retropubic MUS. Its first series defined the two complications **unique to MUS**: mesh penetration of the urinary tract and vaginal mesh exposure.
- Delorme's transobturator route (2001) avoids bladder, bowel and vessel injury. The first single-incision mini-sling (SIMS) was approved in 2006 (TVT-Secur, withdrawn).

Guideline status (American Urological Association/Society of Urodynamics and Female Urology, AUA/SUFU)

- MUS for the index patient is a **strong recommendation**, grade A. Either route is conditional, grade A; SIMS is conditional, grade B.
- Mesh-specific risks **and** non-mesh alternatives must be discussed before a synthetic MUS (Clinical Principle).

2 · Preoperative assessment and patient selection

Why this matters: SUI affects quality of life, not survival, so the work-up confirms SUI and asks whether more tests would change the plan.

- P3 SUI: evaluation in females, role of UDS, and surgical therapy in males — *AUA/SUFU minimum evaluation (history, exam, cough stress test supine then standing, PVR, urinalysis); indications for UDS;*

ValUE; ISD cut-offs

Not in the paper answer:

- **Baseline urgency** predicts a poorer sling outcome. Bother and goals drive the decision.
- **Cystoscopy** is NOT part of the index evaluation. Do it for haematuria, suspected neoplasm or foreign body, or prior pelvic surgery. Upper-tract imaging and MRI are used only when indicated.
- Unexplained haematuria or pyuria also calls for further evaluation.
- ValUE criticism: most screened women were ineligible; surgeons were fellowship-trained.
- Detrusor overactivity (DO) may be stress-induced (urine entering the proximal urethra triggers it), so it can improve after a sling.
- Q-tip movement **>30°** on straining defines hypermobility.
- Further evaluation, including urodynamic studies (UDS), is needed for unclear diagnosis, neurogenic bladder, urgency-predominant mixed urinary incontinence (MUI), high post-void residual (PVR), POP-Q (Pelvic Organ Prolapse Quantification) stage ≥3 without SUI on reduction, or prior surgery.

Urodynamic studies — the controversy

- UDS may be **omitted** in the index patient with demonstrable SUI (conditional, grade B), based on **ValUE**.
- An MUCP **<20 cm H_2O** defines type 3 SUI, that is **ISD**. An abdominal or Valsalva leak point pressure (ALPP/VLPP) **<60 cm H_2O** also means ISD; a value **>90** means little or no ISD.
- ISD is a **matter of degree** (all SUI has some) and now largely a clinical diagnosis. Low MUCP or VLPP predicts failure inconsistently (mainly after transobturator slings). ISD matters most with a **hypomobile urethra**.
- **DO** is the most consequential bladder finding; it may improve or worsen, so consider treating it first. Low capacity and poor compliance worsen outcomes. Detrusor underactivity and Valsalva voiding predict voiding difficulty but should **not exclude** surgery.
- Offer non-surgical options to all: weight loss (dose-related benefit), **pelvic floor muscle training (PFMT)** before surgery for most (adherence limits it), pessaries and bulking *see Ch 120*.

3 · Choosing the sling

Why this matters: mobility, ISD, prior mesh and attitude to mesh decide the sling.

Situation	Preferred option	Reasoning
Index SUI, mobile urethra	MUS (retropubic or transobturator); SIMS acceptable	Equal efficacy, least morbidity
ISD with mobile urethra	Retropubic preferred over transobturator	Meta-analyses favour retropubic
Fixed, immobile urethra	Retropubic MUS, autologous PVS, bulking, adjustable sling	AUA Expert Opinion; mobility predicts MUS success
Recurrent SUI after failed MUS	Retropubic MUS or autologous PVS	Retropubic better than transobturator for recurrence
Mesh complication, urethral injury, fistula, diverticulum, reconstruction	**Autologous PVS**	No mesh near a damaged urethra; adds tissue
Declines mesh	Autologous PVS or Burch *see Ch 116*	—

Situation	Preferred option	Reasoning
Neurogenic SUI on clean intermittent catheterisation (CIC)	Autologous PVS (may be tight)	Only once UDS confirm safe capacity and compliance
Concomitant pelvic organ prolapse (POP) surgery	MUS, PVS or Burch	Conditional, grade C

4 · Pubovaginal sling

Why this matters: the autologous PVS is the fall-back for every difficult case.

- **Indications:** primary or recurrent SUI from **hypermobility and ISD**, stress-predominant MUI, anteroapical POP, neurogenic SUI, **urethral reconstruction** (trauma, mesh perforation, hypospadias, interposition at fistula or diverticulum repair), and failed Burch or MUS.
- **Autologous fascial PVS is the gold standard for all forms of SUI.**
- It gives **dynamic compression without obstruction**. Too much tension causes ischaemia, retention and erosion. Give-way is prevented by **incorporation into the endopelvic fascia with fibrosis**.
- Width **<2 cm** compresses without a constricting band.

Material	Behaviour	Status
Autologous — rectus fascia, fascia lata	Best ingrowth and neovascularisation, minimal reaction; negligible erosion	**Gold standard**; cost is harvest pain and wound complications
Allograft	Processing weakens it; variable, less durable	Mostly withdrawn
Xenograft (porcine, small intestinal submucosa (SIS), bovine)	Weaker, encapsulated; cure falls with time; porcine inferior in trials	Mostly withdrawn
Synthetic at bladder neck	Urethral perforation **15×** and vaginal exposure **14×** other materials (especially woven polyester)	**No longer used at bladder neck**
Vaginal wall sling	Weak; inclusion cysts, vaginal shortening	Historical

> **KEY POINT**
> **Fibrosis paradox**: more fibrosis should mean better support, but the inflammation that drives it also degrades the sling and causes erosion. Autologous fascia remodels with minimal inflammation, which is the ideal balance.

- **Rectus or fascia lata:** strength and cure are equal. Fascia lata avoids abdominal wound problems and bleeds less, but needs repositioning and causes thigh pain. Use fascia lata after **ventral hernia repair**. Narrow (1.5 cm) strips prevent thigh muscle herniation, and a Crawford stripper reduces pain.

5 · Autologous PVS — operative procedure

Why this matters: most PVS problems come from technique, especially tension.

- **Counsel** about voiding dysfunction and de novo urgency; teach CIC in selected women; the sling will not fix bowel symptoms.
- Give one dose of a first- or second-generation cephalosporin (cefazolin) and intermittent pneumatic compression (IPC) with or without heparin. Lithotomy nerve injury (lateral femoral

cutaneous, femoral, peroneal) is rare and almost always recovers.

ALGORITHM

1. Through a 6–7 cm Pfannenstiel incision, mark and excise a **2 × 8 cm** rectus graft.
2. Close the fascia with No. 1 polydioxanone (PDS).
3. Clean the graft, stitch No. 1 PDS to each end and keep it in saline.

- Keep a transverse incision **≥2 cm above the symphysis** for tension-free closure; if impossible (rare), bridge with mesh or biologic.
- **Fascia lata:** incise above the patella over the iliotibial band and strip proximally; graft 8 cm (classically 20 × 2 cm). The **fascia lata is not closed**; compression wrap for 8 h.
- **Vaginal part:** make an **inverted-U incision** with thick flaps. With the **bladder empty**, perforate the **endopelvic fascia** toward the ipsilateral shoulder and finger-dissect the retropubic space. At reoperation, dissect sharply, hugging the pubic periosteum.

ALGORITHM

1. With the bladder empty, pass Stamey needles **from above**, on the back of the pubis, into the vaginal incision.
2. Perform **cystoscopy (70° lens)**. If there is an injury, repass **more laterally** and re-scope after each pass.
3. Confirm ureteric efflux, centre the graft and tag its ends above.

- Do POP repair **before tensioning** and **remove the weighted speculum**.
- **Tension:** tie the sutures over the rectus with **two fingerbreadths** of slack, deliberately **loose**. Tighter only if lifelong CIC is planned.
- Anchor the graft to periurethral tissue (two 4-0 polyglactin); close the vagina watertight; optional pack with or without oestrogen cream.
- Postoperatively: liberal fluids; pack and catheter out with a voiding trial on day 1; if it fails, Foley or CIC and retry within 5 days. No lifting >10 lb and no intercourse for **6 weeks**.

6 · PVS outcomes

Why this matters: PVS rarely fails for stress leakage; "failure" is usually urgency.

- Results are durable even in complex patients (radiation, spinal cord injury, trauma). Cure stays high even with MUCP <4 cm H_2O, so **severe ISD is not a contraindication**.
- **Recurrent SUI:** a repeat PVS, or a PVS after failed MUS (with partial mesh excision), succeeds in most.
- The commonest "failure" is **storage symptoms or urgency urinary incontinence (UUI)**, not stress leakage. De novo UUI occurs in 2–22%. Prior POP surgery predicted de novo urgency; tight technique, age, body mass index (BMI) and radiation did not.
- The four-quadrant modification has high cure; all failures were from storage symptoms.
- Fascia lata and rectus are equal at 1 year. Rectus has more wound problems; fascia lata bleeds less but causes transient thigh pain and numbness. Short (4 × 2 cm) grafts work in ISD.
- **Cochrane, PVS versus MUS:** PVS is probably no better, maybe less effective; MUS is shorter, with fewer complications (except bladder perforation) and less voiding and storage dysfunction.

- A damaged urethra (after mesh excision or diverticulectomy) does worst, yet PVS still dries most.
- **MUI:** the urgency outcome is unpredictable; anticholinergics alone resolve urgency in only about two-thirds. PVS cure in MUI equals pure SUI. PVS with anterior repair works for grade 3–4 cystocele with SUI.

7 · PVS complications and management

Why this matters: obstruction is the main price of a PVS, and it is managed differently from mesh obstruction.

Voiding dysfunction and bladder outlet obstruction (BOO)

- Incidence is 2.5–35%, **more than Burch** (SISTEr: all outlet procedures in the PVS arm). Retention >4 weeks is ≈8%; **permanent retention is ≤5%** (≈2% if not overtensioned).
- The usual cause is **excess tension**. Host factors: poor contractility, dysfunctional voiding and **straining to void**, which tightens the dynamic outlet.
- It presents as weak stream, retention, **storage symptoms only** (urgency commoner than retention), recurrent urinary tract infection (UTI) or pain, even with a low PVR. Exclude new POP kinking the urethra.
- Prediction is unreliable (PVR >100 mL and maximum flow rate (Qmax) ≤20 mL/s are inconsistent), so low detrusor pressure means **counsel about CIC**, not refuse surgery.
- Evaluate by examination (angulation, fixed urethra), PVR (no cut-off), cystoscopy and selective video-UDS; high-pressure, low-flow is often **absent in women**.

> **KEY POINT**
> **The key diagnostic criterion is the temporal relationship** between surgery and symptoms, not a urodynamic number. Preoperative UDS does not predict obstruction (SISTEr).

> **ALGORITHM**
> 1. Most transient retention resolves within **10 days**: use catheter or CIC, timed voiding, PFMT, anticholinergic.
> 2. Autologous obstruction often improves, so wait up to **3 months**.
> 3. If retention is complete, act earlier: at **<6 weeks**, loosen the sling under anaesthesia by downward urethral traction.
> 4. If **>6 weeks** or conservative care fails, then do sling incision or urethrolysis.

- Downward loosening is **not for synthetic slings**. Dilation risks fibrosis. **Transurethral bladder-neck incision fails** (the sling is extraluminal) and risks sphincter damage.

Procedure	Features
Sling incision (transvaginal)	84–100% success; least morbid, so **first choice**; ≈20% recurrent SUI, not prevented by interposition
Transvaginal urethrolysis	Least effective for PVS: cannot relieve suburethral compression
Suprameatal urethrolysis	Better access to the lateral wings
Retropubic urethrolysis	Best results; after failed transvaginal attempt

- Urethrolysis succeeds in 65–93%, with recurrent SUI in 0–19%; prior urethrolysis is the only suggested adverse factor. Failure is usually **inadequate lysis**; a repeat relieves ≈90%.

- **Storage symptoms persist** in over half: use an antimuscarinic or β3-agonist, **sacral neuromodulation (SNM)** (test-phase response predicts benefit), or onabotulinumtoxinA 100 U (≈40% dry at 12 weeks).
- Counsel: lysis may not cure urgency and may make a dry woman leak; untreated female BOO has unknown long-term effects.
- Recurrent SUI after lysis: bulking, repeat PVS or MUS.

Perforation, exposure and other complications

- Perforation is rare with fascia (missed needle injury, excess tension, instrumentation). Urethral perforation needs **sling excision and repair** (with or without Martius flap); bladder erosion is removed endoscopically. Bladder-neck sling perforations present at 1–18 months (mean ≈9).
- An exposed synthetic PVS is removed; recurrent SUI (44–100%) is treated with a second, **autologous** PVS.
- Mortality is ≈**3 per 10,000** anti-incontinence procedures; harvest-site **incisional hernia** ≈1%.

8 · Mid-urethral sling — mechanism and materials

Why this matters: why a loose tape works explains why a fixed urethra fails.

- The MUS **secures the posterior urethral wall**, with or without compression and dynamic **kinking**. Retropubic tapes anchor in endopelvic fascia, transobturator tapes in obturator muscles and membrane. Mesh is fixed along its **whole length**, so 50–93% stay dry after midline incision.
- Preoperative **immobility predicts failure**. Postoperative mobility is harmless; ideal is a loose sling with a mobile urethra. **Exception, SIMS:** success needs restricted mobility, so it is set tighter.
- Whether MUS obstructs is disputed; most surgeons tension both routes alike, rarely with an objective test.

> **KEY POINT**
> **Why does a loose tape work?** At rest it does not touch the urethra, so voiding is unobstructed. On coughing the mobile urethra descends onto the tape; the posterior wall is arrested, the urethra kinks and closes. A fixed urethra never meets the tape, and tightening the tape to compensate obstructs.

- The material is **type 1 (Amid) macroporous monofilament polypropylene** with pores **>75 µm**, which admits macrophages and fibrovascular ingrowth. UraTape and ObTape (<50 µm) had high exposure and were withdrawn.

9 · MUS operative procedures

Why this matters: each route has its own danger structures.

- P2 P1 Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT; TOT in SUI — *TVT bottom-up steps, TOT outside-in/inside-out steps, tensioning, Cochrane complication rates*

Not in the paper answer:

- Document counselling about the **permanent implant**, mesh-specific complications, voiding and storage dysfunction, and non-mesh alternatives.

- Cough-test tensioning under local is unnecessary (same outcomes under spinal). A single antibiotic dose is still given.
- Bottom-up beats top-down: better subjective cure, less voiding dysfunction, perforation and exposure.
- TVT-O exits 2 cm above urethral level and 2 cm lateral to labial folds; winged guide at 45°, helical passer. Outside-in possibly injures the urinary tract more.
- Brief thigh flexion (≥70°) helps trocar passage.

ALGORITHM

1. Infiltrate local; make two suprapubic stabs ≈2 cm from midline.
2. Make a 1.5 cm midline vaginal incision 1.5 cm from the meatus.
3. Make minimal lateral tunnels to the endopelvic fascia, **not perforating** it (unlike PVS).
4. The catheter guide deflects the bladder to the opposite side.
5. Pass the trocar through Retzius and rectus, **hugging the back of the pubis**; repeat on the other side.
6. Perform cystoscopy.
7. Tension loosely (clamp between tape and urethra), remove sheath, trim.

- TVT kit: two 5 mm trocars, 40 × 1.1 cm polypropylene tape. Local: ≈5 mL at vaginal and suprapubic sites, 20 mL behind the pubis, 10 mL each side of the urethra.
- Keep the trocar in **metal-to-bone contact** with the back of the pubis, keeping it extravesical and extraperitoneal.
- **Transobturator, outside-in (TOT):** hips **hyperflexed (120°)**; dissect laterally to the ischiopubic ramus; puncture the labiocrural fold at the adductor longus notch. The trocar pierces the obturator membrane ("give") and turns around the ramus onto the finger. Check the vaginal sulcus for a button-hole.
- **Inside-out (TVT-O)** has the same cure, with more voiding dysfunction and fewer vaginal perforations.
- **SIMS** uses only a vaginal incision and anchors in **obturator internus**; set it **tighter** (haemostat tip passes), close to the urethra, with wide, deep tunnels.

Feature	Retropubic (TVT)	Transobturator	SIMS
Path / anchor	Retzius, rectus fascia; U-shape	Obturator foramen; flatter hammock	Obturator internus only
Cure (index SUI)	Equal to transobturator up to 5 y; best long-term data (>17 y)	Equal; slightly more repeat surgery	Non-inferior to transobturator (TVT-Secur excluded)
ISD / recurrent SUI	**Preferred**	Worse with low VLPP	Limited data
Bladder perforation	**4.5%**	**0.6%**	Rare
Vascular/visceral injury, bleeding, voiding dysfunction	More	Less	Less
Pain	Suprapubic 2.9%	**Groin 6.4%** vs 1.3% retropubic	Least
Vaginal exposure	About 2%	About 2%	Similar (excluding TVT-Secur)

Feature	Retropubic (TVT)	Transobturator	SIMS
Cystoscopy	Mandatory (70°)	Recommended; check the urethra	Essential (30° suffices)

10 · Intraoperative cystoscopy

Why this matters: trocar injury must be found before mesh is left.

- P1 Management of perforated mid urethral sling — *70° lens, full bladder; bladder injury: repass laterally and continue; urethral injury: no mesh, staged repair*

Not in the paper answer:

- Inspect the urethra too after transobturator and SIMS passes.
- After a PVS, also confirm ureteric efflux.
- A bladder injury is benign: leave a catheter overnight to 72 h.

11 · MUS outcomes

Why this matters: these are the figures quoted when counselling.

- Retropubic slings are the most studied (17–20 years); objective cure stays high, subjective lower, and <85% on strict definitions. Cochrane (81 trials): retropubic and transobturator cure are equal from short to long term.
- TVT equals colposuspension at 5 years; de novo urgency <2% vs <5%; later POP commoner after Burch.
- In the Trial of Mid-Urethral Slings (TOMUS) at 5 years, satisfaction stayed high (79% retropubic, 85% transobturator) despite falling objective success.
- Long term, **de novo urgency rises** (≈⅓ at 10–20 years); retention is 0–7%. A one-step fall in leak frequency satisfies most.
- In Cochrane SIMS did worse than retropubic, driven by **TVT-Secur (withdrawn)**; others about equal transobturator, with less pain. Failure fell from 20% to 8% when sited closer to the urethra. The key long-term endpoint is **reoperation**.

Group	Key points
MUI	Urgency resolves in ≈60–75% (best when stress-predominant). De novo urgency risk rises with age, obesity, nocturia, DO and low capacity. Persistent DO: transobturator < retropubic < PVS. New storage lower urinary tract symptoms (LUTS) may mean BOO
ISD	About 50% cure with urodynamic ISD. An immobile urethra predicts failure more strongly than a low VLPP, and low VLPP is not a contraindication to retropubic MUS. Transobturator: 12× odds of persistent SUI with low VLPP
Recurrent SUI	No level 1 evidence; repeat MUS is commonest, retropubic better than transobturator. Cure ≈81–90% but below primary surgery (≈62% vs 86%); more urgency, and more bladder perforation after prior retropubic surgery
Concomitant POP	A concomitant sling lowers postoperative SUI and reoperation but raises voiding dysfunction. About 30% of SUI resolves after POP repair alone. A prophylactic sling only after counselling (OPUS); in one review only the retropubic subgroup benefited
Elderly	Safe; similar or lower cure, more urgency and voiding dysfunction. Q-tip >30°: 90% continent; <30°: 42%; <10°: 80% stay incontinent. Cure ≈64% at ≥75 y vs 89% at 55–64 y. **Frailty, not age**, predicts readmission. Do not exclude by age (AUA Expert Opinion)
Obese	Success >80% but lower (≈87% vs 66% at 5 y). Obesity is a risk factor for TVT failure, worse with BMI >35; more wound infection, fewer bladder punctures

12 · MUS complications and management

Why this matters: most MUS complications are minor, but a few need prompt action.

KEY POINT

Concurrent or staged sling at POP repair. Concurrent: one operation and less SUI (≈39% falls to 8–19%), but more obstruction, storage symptoms and mesh events; tensioning is hard if SUI is only occult. Staged: avoids an unnecessary sling but commits a third to a second operation (usually within 12 months). Decide after testing for occult SUI with the prolapse reduced.

- Bladder trocar injury is ≈2–4%, almost all retropubic; risk rises with prior retropubic surgery and thin habitus. Vascular (≈0.07%) and bowel (≈0.04%) injuries are rare, but deaths have been reported.
- †Unexplained hypotension, falling haemoglobin, pain, ileus or sepsis after a retropubic sling needs urgent imaging or exploration.
- Bleeding is 2–3%, more retropubic (≈4% bled ≥300 mL needing intervention in one series). Ultrasound shows retropubic haematoma >5 cm in ≈1 in 8, usually **asymptomatic, so observe**.
- **UTI is the commonest complication** (3–4% at 30 days; ≈1 in 4 at 1 year). Risks: age, diabetes, BMI >40, prolonged catheter. It peaks in the first month (day 7), is less after transobturator slings, and marks a worse course (more reoperation, de novo UUI).
- Vaginal wound infection is 0.7–2%; rarely necrotising fasciitis, abscess or **adductor myositis** (may present years later).
- **Groin pain** after transobturator slings affects 16–26% early, ≈4% at 12 months. Treat with non-steroidal anti-inflammatory drugs (NSAIDs), physiotherapy, trigger-point injection; if refractory, excise: **complete removal** relieves more than partial.

Voiding dysfunction and retention

- Inefficient but volitional voiding: timed voiding, biofeedback, PFMT and selective drugs first.
- The tape is **too tight** or **too proximal** (obstructing slings usually lie near the bladder neck), or POP kinks the urethra (even over a normal sling ≥10 years later). It is commoner after retropubic slings (TOMUS: 2.7% vs 0%). Evaluate by **temporal relation**, examination, PVR, cystoscopy and selective UDS.

ALGORITHM

1. Retention or high PVR: CIC or catheter; most resolve within 2 weeks.
2. If obstruction persists, perform a **transvaginal midline sling incision at 2–4 weeks**.
3. Beyond 3 months the tape is fixed, so excise its lateral portion.
4. Formal urethrolysis is rarely needed.

- **Urethral dilation is harmful; do not perform it**. After incision 70–100% stay continent. **Early** release gives better voiding and **less** recurrent SUI; delay is the main predictor of recurrence.
- "Recovered" women often keep frequency and urgency, suggesting partial obstruction.

KEY POINT

Why is the timing different from PVS? Fascial obstruction often settles, so one waits up to 3 months. Mesh is fixed by ingrowth and rarely loosens; late release needs wider dissection and causes more recurrent SUI, so release within ≈4 weeks.

Storage symptoms, exposure and sexual function

- De novo overactive bladder (OAB) is ≈6–12% (up to 15–30%), equal for all MUS types; predictors are age, DO, low capacity and high detrusor pressures. A clear temporal link means **partial BOO first**; **cystoscopy is mandatory**. Treat with behavioural measures and drugs (less effective than in idiopathic OAB), then sling revision or SNM, percutaneous tibial nerve stimulation (PTNS) or botulinum toxin.
- Vaginal exposure is ≈2%; †it causes discharge, bleeding, dyspareunia or partner discomfort. †Small: topical oestrogen with or without excision; larger: excision. Urinary tract mesh: excision and repair, no immediate new mesh *see Ch 119*.
- De novo dyspareunia is ≈3–14%, tracking with de novo urgency and incomplete emptying more than mesh; sling removal may relieve it. Many improve as **coital incontinence** stops; in the TVT–colposuspension trial dyspareunia fell from 34% to 13% at 5 years.

13 · Adjustable slings

Why this matters: a niche option for complex SUI.

- **Remeex** is mesh on sutures to a suprapubic **varitensor**, adjusted after surgery, for **complex SUI** (ISD, fixed urethra, reoperation).
- Cure is ≈85–90%; complications 32–51%, bladder injury 7–14%, readjustment 14–43%. Detrusor underactivity may cause retention (a quarter wanted removal). Higher BMI and pad weight predict worse results. No level 1 evidence.

14 · Assessing the failed or troublesome sling

Why this matters: persistent SUI, obstruction, OAB and mesh problems need different treatment.

ALGORITHM

1. Define the symptom: SUI, storage, voiding, or pain/discharge.
2. Establish the **timeline** in relation to surgery.
3. Examine for exposure, POP, urethral mobility and tenderness over the tape.
4. Check PVR and urine culture, perform **cystoscopy**, and add UDS selectively.
5. Tailor treatment to the cause.

- If SUI recurs with a mobile urethra, repeat a retropubic MUS or do a PVS; if the urethra is fixed or there is ISD, use PVS, bulking or an adjustable sling.
- If storage symptoms have a temporal link, then treat obstruction first. Pain or exposure needs excision (complete for pain) *see Ch 119*.

15 · Mesh controversy and regulatory status

Why this matters: patients ask about mesh warnings; documented counselling is expected.

- The **2010 US Food and Drug Administration (FDA)** safety communication on mesh for POP and SUI was based on reports to the **MAUDE** (Manufacturer and User Facility Device Experience) database. Professional bodies still endorse MUS with documented counselling.
- †The 2019 FDA order stopping transvaginal mesh concerned **POP**, not MUS. The UK (2018) paused MUS use pending safeguards *see Ch 119*.

16 · Landmark trials

Why this matters: these trials underpin the guidelines.

Trial	Design	Result	Take-home
ValUE	Randomised controlled trial (RCT), non-inferiority, 630, uncomplicated SUI: UDS vs office evaluation	Office evaluation non-inferior at 1 y	UDS may be omitted in the index patient
SISTEr	RCT, 655: rectus PVS vs Burch	PVS higher cure (47% vs 38% at 2 y); more voiding dysfunction, UTI and UUI; all BOO surgery in PVS arm; persisted at 5 y	PVS more effective but more obstructive
Ward–Hilton	RCT: TVT vs open colposuspension	Equal to 5 y; TVT more bladder injury; Burch more morbidity and POP	MUS replaced Burch
TOMUS	RCT, 597: retropubic vs transobturator	80.8% vs 77.7% (1 y, equivalent); surgery for voiding dysfunction 2.7% vs 0%; †more groin and neurological symptoms with transobturator; 5 y 51% vs 43% (equivalence not met)	Either; retropubic slightly more durable, more voiding dysfunction
OPUS	RCT (327): POP repair with or without prophylactic retropubic MUS in stress-continent women	Incontinence at 12 months 27% vs 43%; number needed to treat (NNT) 6.3; more perforation, UTI and bleeding	Reduces SUI but adds risk, so counsel

117 Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

1 · Goals and principles

Why this matters: prolapse is a quality-of-life (QoL) problem, so operate for symptoms and bother.

- P2 Vaginal prolapse repair — *QoL condition, first operation best chance, restorative/compensatory/obliterative, vaginal (less morbid) vs abdominal (more durable), apex is the cornerstone*

Not in the paper answer:

- **Success means absence of symptoms** (no bulge), which drives satisfaction more than anatomy; satisfaction tracks her own goals.
- Routine events (catheter, pain, constipation, urinary tract infection [UTI]) are perceived as complications. Preoperative education improves the Patient Global Impression of Improvement (PGI-I) without changing objective cure.
- **Consent sequence:** alternatives, purpose and limits, benefits, risks, hospital course, home course, coping with a catheter.
- Obliterative surgery is contraindicated if intercourse is desired. Open and laparoscopic routes have equal complications; laparoscopic recovery is faster.
- Decide on pelvic organ prolapse (POP) surgery from grade, symptoms and bother together; declining surgery is a legitimate choice.
- Individualise to physiological age, comorbidity, prior surgery and sexual activity. **The first operation is the best chance**: planes are then lost and recurrence rises with each re-attempt.
- Compensatory (graft) repair is for weak tissue or a failed repair. POP is usually multicompartmental.

2 · Choice of procedure by compartment

Why this matters: each Pelvic Organ Prolapse Quantification (POP-Q) point maps to a compartment and its operations.

- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *POP-Q points (A distal, B proximal, D only with cervix), DeLancey levels, operation by defect, SSLF and promontory anatomy, uterus-preserving options*

Not in the paper answer:

- Enterocele (point D) is closed by culdoplasty: **McCall** vaginally; **Halban** or **Moschcowitz** abdominally.
- Also omitted: retropubic urethropexy (Aa), colpoperineopexy (Ap) and wedge colpectomy (Ba).

Compartment	POP-Q point	Organ
Anterior	Aa / Ba	Urethrocele / cystocele
Middle	C / D	Cervix or cuff / enterocele
Posterior	Ap / Bp	Rectocele / enterocele; perineal body

Site	Vaginal	Abdominal
Aa — urethra	Anterior repair; bladder neck suspension; sling	Retropubic urethropexy

Site	Vaginal	Abdominal
Ba — bladder	Anterior repair; paravaginal; colpocleisis; wedge colpectomy	Paravaginal; sacrocolpopexy
C — cervix/cuff	Uterosacral ligament suspension (USLS); iliococcygeus; sacrospinous ligament fixation (SSLF); Manchester; hysteropexy; vaginal hysterectomy; colpocleisis	Hysterectomy; USLS; sacrocolpopexy; uterine suspension
D — cul-de-sac	**McCall** culdoplasty	**Halban**, **Moschcowitz** culdoplasty
Ap — posterior	Posterior repair; site-specific repair	Colpoperineopexy

KEY POINT

Only paravaginal repair and uterosacral suspension are done by both routes. McCall is vaginal; Halban (linear sutures) and Moschcowitz (purse-string, which may angulate the ureter) are abdominal.

3 · Anterior compartment

Why this matters: anterior repair has the highest recurrence, so know its limits.

- A defect may be **central** (midline thinning of the pubocervical fascia) or **lateral/paravaginal** (detached from the arcus tendineus fasciae pelvis, ATFP). Most are combined, often with apical loss. **Anterior repair does not suspend the apex.**

ALGORITHM

1. In lithotomy, catheterise, hydrodissect and incise in the midline from apex to bladder neck.
2. Dissect the epithelium off the pubocervical fascia laterally to the ATFP and reduce the bladder.
3. Plicate with interrupted 2-0/3-0 delayed-absorbable sutures from bladder neck to apex.
4. Trim judiciously and close; give intravenous (IV) dye, do **cystoscopy** and pack.

- A mid-urethral sling (MUS) needs a separate incision; a pubovaginal sling (PVS) uses one incision extended to the mid-urethra. Deep sutures may enter or kink bladder or ureter, so completion cystoscopy is mandatory.
- Anterior repair is abandoned for stress urinary incontinence (SUI): American Urological Association (AUA) meta-analysis failure approached **40%**.
- Recurrence depends on the **definition**: 30–46% for "POP-Q ≥2" versus 9–23% for "beyond hymen"; objective recurrence often leaves satisfaction intact.
- Absorbable mesh gives a modest gain; biological grafts add little. Polypropylene lowers recurrence and reoperation versus native tissue but causes far more graft complications, without better QoL or dyspareunia. Hence the Food and Drug Administration (FDA) 522 orders and **withdrawal of transvaginal synthetic POP mesh** (US 2019) *see Ch 119*.
- A concomitant sling **halves anterior recurrence** (55% lower), more than any graft.

Complication	Point / management
De novo SUI	Rises with stage: stage 2 **41%**, early 3 **53%**, advanced 3–4 **66%**; point Ba predicts it
Detrusor overactivity	De novo in 5–7%; pre-existing overactivity resolves in up to 63%

Complication	Point / management
Retention	Usually from the sling: clean intermittent catheterisation (CIC); if persistent, sling incision or urethrolysis. Hypocontractile detrusor: CIC, then neuromodulation, then suprapubic catheter (SPC)
Bleeding (>350 mL)	Vaginal vascularity, venous complex under pubis; bleeding means wrong plane; worse at redo
Bladder injury	Drain bladder before perforating endopelvic fascia; two-layer closure; Martius if irradiated
Others	Ureteric obstruction, vesicovaginal fistula (VVF), vaginal shortening, inclusion cyst, dyspareunia

KEY POINT

Confirming ureteric efflux: indigo carmine (blue, IV), fluorescein 10% (green, IV) or phenazopyridine (orange, oral). No efflux: ureteric catheter or retrograde study, or remove plication sutures. Under-hydration delays efflux, but never close without proving patency.

Paravaginal repair

- Reattach the pubocervical fascia to the ATFP (White vaginal; Richardson abdominal) with non-absorbable sutures 1 cm apart: **5–7** vaginal, **4–6** abdominal; then cystoscopy.
- Lateral recurrence is 2–8%, but central recurrence is ~**22%**, so these are independent failures. Abdominal is more durable.
- **Neither colporrhaphy nor paravaginal repair treats SUI.** The vaginal route has more serious complications (transfusion 12%, haematoma, abscess, bilateral ureteric obstruction).

4 · Occult SUI and the concomitant sling

Why this matters: correcting prolapse can unmask SUI, so decide on a sling beforehand.

- High-stage anterior POP kinks the urethra and masks SUI; "stress incontinence on prolapse reduction" occurs in 8–66% of continent women.
- Test **with prolapse reduced** (pack, swabs, speculum, pessary); no method is standardised; sensitivity is low, specificity high.
- **Staged** argument: sling obstruction needing intervention (8.5%) equals intervention for SUI without a sling (8.3%).
- The **OPUS calculator** uses 7 predictors (accuracy 0.72 vs 0.54 for the stress test). **CUPIDO** (9 predictors): urodynamics add nothing in asymptomatic women. A six-predictor model is similar.
- With subjective SUI and native repair or colpocleisis, <50% have bothersome SUI, so the sling may be omitted. SUI symptoms with negative tests: ~30% later need intervention.

KEY POINT

A sling also helps the prolapse: distal anterior support makes the cystocele repair last. **Why not everyone?** In OPUS it caused more bladder perforation, UTI, bleeding and incomplete emptying, with unchanged pelvic floor scores. Weigh per patient.

ALGORITHM

1. In advanced anterior POP, do a reduction stress test, with or without urodynamic studies (UDS).
2. If SUI is symptomatic, occult or urodynamic, then offer a concomitant procedure.
3. If she is continent and the test is negative, then weigh individual risk and benefit (OPUS calculator).
4. The sling is **selective, not routine**.

5 · Apical compartment

Why this matters: the apex is the cornerstone; failing to secure it raises recurrence exponentially.

Uterosacral ligament suspension (USLS)

- USLS preserves the natural axis; the ligament breaks at discrete points.
- **The ureter lies 4 cm from the ligament at the sacrum but only 0.9 cm at the cervix**. Sutures catch the ureteric sheath, so injury is **kinking**, not ligation. Sutures near the sacrum entrap S1–S4 (buttock and posterior thigh pain); S3 lies >4.6 cm cephalad of the spine.
- The **safe zone** is the intermediate portion, ~1 cm posterior, at the level of the ischial spine; pass needles **lateral to medial**.

ALGORITHM

1. After vaginal hysterectomy, place cuff stays at 3 and 9 o'clock, excise the enterocele sac, pack bowel and palpate the spines.
2. Place Allis clamps at 5 and 7 o'clock and pass **2–4** non-absorbable sutures per side.
3. Do any anterior repair or sling now.
4. Bring the sutures through the apical pubocervical and rectovaginal fascia and tie.
5. Perform **cystoscopy after tying, before trimming**. If there is no efflux, remove the most lateral ipsilateral suture first.

- Confirm the ureter lies laterally before suturing (the relation shifts with prolapse); stent if hard to palpate. Small bowel injury and abscess are rare; voiding and SUI usually improve.
- Versus SSLF, USLS risks pudendal and gluteal vessels less, keeps the midline axis and closes the cul-de-sac. The abdominal variant uses 3 permanent sutures per side (Shull).
- Apical success is ~98% and anterior ~81%; ureteric kinking or injury occurs in **1–11%**. Absorbable sutures are non-inferior; premenopausal women recur more; de novo dyspareunia can occur.
- **OPTIMAL**: USLS equals SSLF (~65% at 2 years on a strict composite), and perioperative pelvic floor muscle training (PFMT) adds nothing.

Sacrospinous ligament fixation (SSLF)

- SSLF (Richter, Nichols) is extraperitoneal and preserves length, by posterior or anterior approach. The 7–8 cm ligament lies under coccygeus; the pudendal bundle runs round the spine, inferior gluteal vessels behind.
- The **safe zone** is **1.5–3.0 cm medial to the ischial spine**, into the ligament, at limited depth. Too lateral risks the pudendal bundle; too medial hits weak ligament and sacral nerves.

ALGORITHM

1. Mark the apex, incise in the midline, enter the perirectal space (rectum medial) and palpate spine and ligament.
2. Place retractors and pass 2 sutures (Capio or Miya hook).
3. Complete other repairs, then attach the sutures to the apex (pulley stitch if permanent).
4. Close the vagina halfway and tie the apex flush, with **no suture bridge**.
5. Do cystoscopy and close.

- SSLF is usually right-sided (2 sutures 2 cm apart); bilateral or anterior approaches are more midline, with anterior recurrence ~7–9%. The **pulley stitch** keeps the knot out of the lumen.
- Cure is ~85% and apical recurrence ~5%. Stage 4 predicts recurrence; with no defect at 6 weeks only 3% are reoperated.
- **Drawback:** the axis is deflected posteriorly and right, causing **anterior recurrence** (~18%). Concomitant anterior repair helps only when Ba is > +1 (with SSLF, not USLS).
- Gluteal pain occurs in **15%** and resolves in 2–3 months (local anaesthetic injection if needed). Also pudendal injury, rectal injury (repair primarily), haemorrhage.

KEY POINT

Both ligaments are safest in their **middle third** and dangerous at their bony ends: the intermediate portion for USLS, and 1.5–3 cm medial to the spine for SSLF.

Iliococcygeus suspension is for deficient uterosacrals: bilateral sutures into iliococcygeus fascia **1 cm distal to the ischial spine**; anterior recurrence is commonest.

Sacrocolpopexy (SCP)

- **Indications:** other defects (including SUI) repairable at one sitting, failed vaginal repair, young sexually active women, isolated vault prolapse, or wish for the most durable repair. It maximises vaginal length without distortion.
- Use **type I macroporous monofilament polypropylene** with secure fixation. Fascia lata recurs in ~1/3 versus <1/10 with polypropylene.

ALGORITHM

1. Low lithotomy with a vaginal stent or EEA (end-to-end anastomosis) sizer; sigmoid to the left.
2. Incise the peritoneum over the promontory and along the **right** of the rectum (watch the right ureter) down to the anterior longitudinal ligament.
3. If there is a large enterocele, then add a Halban culdoplasty.
4. Dissect the bladder off the anterior vagina (6–8 cm) and the rectum off the posterior wall.
5. Attach a Y-mesh to both walls **without entering the lumen**, tension it, and fix it with 2–3 permanent sutures at **S1**.
6. **Retroperitonealise** the mesh, add an anti-incontinence procedure if indicated, and perform cystoscopy.

- Too low (S3–4) risks the presacral venous plexus; too high risks middle sacral vessels and the **left common iliac vein** (often medial to the artery).
- Tension: open, two-finger gap between graft and rectum; minimally invasive, extend the vagina with the sizer then release. Avoid hypersuspension (pain, de novo SUI). Absorbable mesh-to-vagina sutures are equivalent.

- Robotic: ports ≥9 cm apart, steep Trendelenburg, side-dock; extra time is suturing. If the vagina is entered, close in two layers; a wrong plane raises exposure. Sutures or tacks can cause spondylodiscitis.
- Open, laparoscopic and robotic SCP succeed in 87–100%; the robot saves ~90 mL blood but costs more. Learning curve: 30–40 laparoscopic, ~50 robotic.
- Failures are mainly **posterior or distal**, with almost no apical retreatment. Isolated SCP leaves cystocele or rectocele in up to one third; ~20% need vaginal repair, so counsel.

SCP complication	Rate / management
Haemorrhage (presacral)	~4%; †pressure, sutures, bone wax, sterile thumbtacks
Cystotomy / enterotomy / ureter	3% / 1.6% / 1%; bladder is the commonest robotic visceral injury
Ileus / small bowel obstruction	3.6% / 1.1%
Mesh exposure	**~10% by 7 years** (E-CARE); mean onset 14 months; more with microporous mesh (Gore-Tex pores let bacteria in but keep macrophages out); excision *see Ch 119*

- **Hysterectomy at SCP:** total hysterectomy raises exposure, so go **supracervical** or close the cuff meticulously.
- Cochrane, vaginal apical repair vs SCP: vaginal gives more prolapse awareness (relative risk [RR] 2.1), repeat surgery (2.3), recurrence (1.9), SUI (1.9) and dyspareunia (2.5); yet it suits many older women. USLS equals SSLF.

ALGORITHM

1. If she is elderly or comorbid and not sexually active, then offer colpocleisis.
2. If she is fit, has primary prolapse and prefers the vaginal route, then offer USLS or SSLF, with or without vaginal hysterectomy or hysteropexy.
3. If young or sexually active, or with recurrence, short vagina or high stage, then offer SCP (minimally invasive).

6 · Hysterectomy vs uterine preservation

Why this matters: many women want to keep the uterus.

- †Hysterectomy alone does not cure prolapse, so always resuspend the apex.
- At SCP, keeping the cervix lowers mesh exposure. Vaginal hysterectomy with vault suspension is also effective.
- Sacrospinous hysteropexy is non-inferior (newer data: superior) to vaginal hysterectomy with USLS, and to uterosacral hysteropexy.
- †Preservation needs a normal uterus: exclude abnormal bleeding, endometrial or cervical disease and Lynch syndrome, and keep cervical screening up to date.

7 · Obliterative surgery — colpocleisis

Why this matters: for frail older women it is quick, low-risk and effective.

- It suits older women not wanting intercourse; regional or local anaesthesia; low morbidity and recurrence.

- **Total colpocleisis** (after hysterectomy): the whole vagina is denuded and inverted with purse-string or transverse sutures. The enterocele need not be opened; layered closure obliterates it.
- **Le Fort** (uterus in place): anterior and posterior rectangles are excised (from 3 cm below the meatus to 3 cm from the cervix), leaving **lateral channels** for drainage. Transverse rows run from the leading edge to above the levator plate.
- Both need **high perineorrhaphy and levator myorrhaphy** and cystoscopy. If the uterus stays: Pap smear, ultrasound, with or without endometrial biopsy. Hysterectomy removes the cancer and **pyometra** risk but raises complications (odds ratio [OR] ~1.9).
- Spare ≥**1.5 cm** of distal anterior wall (urethral traction causes SUI). Add a sling for SUI or occult SUI; in the frail consider SPC.
- Success is ~90–98% and satisfaction >90%, but regret reaches **13%**. 43% are over 80, so optimisation matters most.

8 · Posterior compartment

Why this matters: bowel symptoms often persist after rectocele repair.

- Symptoms: bulge, splinting or digitation, trapping (**not faecal incontinence**), dyspareunia. First exclude slow transit and **anismus** (biofeedback).
- **Benefit is predicted** by bulge, splinting, incomplete emptying on defecography and a large rectocele, but not by constipation. **Constipation often persists**, and two-thirds of women who digitate may resume.
- **The transvaginal route beats transanal** (recurrence 7% versus 66%).
- **Midline fascial plication** (avoiding wide lateral bites), with or without perineal body repair, is the International Consultation on Incontinence (ICI) procedure of choice. In **site-specific** repair the commonest defect is **transverse** (detachment from the perineal body); recurrence is higher in advanced POP.
- Perineorrhaphy: excise a triangle, approximate the perineal body with mattress sutures and leave a smooth contour, because ridges cause dyspareunia.

ALGORITHM

1. Repair anterior and apical compartments first; hydrodissect and incise in the midline.
2. Dissect the rectovaginal fascia off the epithelium; if the hernia is large, place a purse-string at its base.
3. Plicate in the midline (2-0 absorbable), incorporate the perineal body, trim and close.

- Plication relieves bulge, straining and splinting; satisfaction follows relief of defecation more than POP-Q. Cochrane: evidence insufficient.
- **Levator plication has been abandoned** because dyspareunia reaches 50%. **Grafts** are not supported (complications RR 1.8).

9 · Complications across POP surgery — summary

Why this matters: correct planes, safe zones and cystoscopy prevent most complications.

Problem	Prevention / management
Ureteric injury (USLS highest, 1–11%)	Bites lateral to medial; cystoscopy after tying; remove the most lateral suture; stent; reimplant

Problem	Prevention / management
Bladder/rectal injury	Correct plane, drained bladder, rectal finger; layered closure
Mesh exposure	Type I mesh, full-thickness flaps, supracervical hysterectomy; oestrogen or excision
Recurrence	Anterior after USLS/SSLF, posterior after SCP; most are asymptomatic
Buttock pain	Middle-third placement; observe up to 3 months; nerve injection
De novo SUI / retention	Reduction testing; selective sling; CIC; sling incision

10 · Landmark trials

Why this matters: these trials shape the decisions above.

Trial	Design	Result	Take-home
CARE	Randomised controlled trial (RCT): SCP with or without Burch, continent women, stage 2–4	SUI 24% vs 44% at 3 months	Prophylactic Burch reduces de novo SUI
E-CARE	7-year extension	SUI 62% vs 77%; mesh exposure ~10%; composite failure 34–48%	The benefit is a delay; exposure rises with time
OPUS	RCT: vaginal POP repair with MUS vs sham	27% vs 43% at 12 months; more perforation, UTI, bleeding, retention	Selective sling
OPTIMAL	RCT: USLS vs SSLF, with or without PFMT	Equal (~65%) at 2 y; PFMT no effect	Native apical repairs are equivalent
Bilateral SSLF vs abdominal SCP	RCT	Earlier failure and 3× cystocele reoperation after SSLF	Abdominal more durable
Transvaginal vs transanal rectocele	RCT	Recurrence 7% vs 66%	Repair vaginally

119 Genitourinary Complications of Mesh and Their Repair

1 · Mesh, terminology and regulation

Why this matters: mesh type and correct terms guide consent and treatment.

- The pelvic floor differs from the abdominal wall: it is near viscera, stretches greatly and has abundant microbiota. Harms are pain, exposure, extrusion and injury to nearby structures.

Amid type	Pore	Example	Behaviour
I	Macroporous >75 µm	Knitted monofilament polypropylene	**Only recommended mesh**
II	Microporous ~10 µm	Polytetrafluoroethylene (PTFE, Gore-Tex)	Encapsulated; >30% full-length slings excised
III	Macropores in microporous base	Mersilene, Mycromesh	Extrusion ~20%
IV	Submicronic	Silicone-coated	Extrusion/sinus up to 40%

KEY POINT

Why pore size matters: bacteria (~1 µm) enter pores that macrophages (~50 µm) cannot, so infection is sealed from immunity. Large pores (~1500 µm) allow fibrocollagenous integration. Woven polypropylene (Marlex) is stiff and extrudes. Mesh adds no cancer or autoimmunity.

- P1 Management of perforated mid urethral sling — *IUGA/ICS terms: exposure, extrusion, perforation; "erosion" not used; trocar injury and late perforation management*

Not in the paper answer:

- Endoscopic excision leaves mesh in the wall, which may re-extrude.
- After urethral mesh excision, SUI occurs in 42%; a stricture needs buccal urethroplasty.
- IUGA and ICS (urogynaecology and continence societies) also define contraction (shrinkage) and prominence (folding without epithelial defect).

† **CTS (Category–Time–Site) coding**: Category 1–3 vaginal (1 no separation; 2 exposure ≤1 cm; 3 >1 cm or extrusion), 4 urinary tract, 5 bowel, 6 skin/musculoskeletal, 7 systemic; suffix A asymptomatic, B symptomatic, C infection, D abscess. Time T1 ≤48 h, T2 48 h–2 months, T3 2–12 months, T4 >12 months. Site S1 vaginal suture line, S2 vagina away from it, S3 trocar path, S4 other skin/musculoskeletal, S5 intra-abdominal.

- The US Food and Drug Administration (**FDA**) grades devices class I low, II intermediate and III high risk (premarket approval). Male and female stress urinary incontinence (SUI) mesh and **transabdominal pelvic organ prolapse (POP) mesh are class II**.
- Transvaginal POP mesh had warnings in 2008 and 2011, became **class III (2016)** and stopped selling in **2019**; slings and sacrocolpopexy are exempt.
- Multi-incision midurethral slings (MUS) and sacrocolpopexy are proven safe; mini-slings are not.
- Give written mesh information, standard removal consent and an explant-return policy.

2 · Risk factors and prevention

Why this matters: avoiding mesh in a bad field prevents most complications.

- **Contraindications to vaginal mesh** are simultaneous urethrovaginal fistula repair, urethral injury or mesh perforation, and diverticulectomy. With **intraoperative urethral injury, abandon mesh** and place a sling later.
- Relative risks: prior fistula or diverticulum surgery, radiation, poor or oestrogen-deficient tissue, smoking, obesity, age extremes; in men, radiotherapy, urethrotomy.
- POP mesh risk rises with high mesh volume, **shallow vaginal dissection**, hysterectomy, diabetes and sexual activity.

3 · Evaluation

Why this matters: know which device was used and where.

> **ALGORITHM**
>
> 1. History and **prior operative notes**.
> 2. Speculum and pelvic exam: exposure, bands, point tenderness, levator hypertonicity.
> 3. If haematuria, recurrent urinary tract infection (UTI), storage lower urinary tract symptoms (LUTS) or voiding pain, then cystourethroscopy.
> 4. If voiding dysfunction, then urodynamics (UDS).
> 5. Translabial ultrasound for a sling; **MRI** for sacrocolpopexy mesh (invisible on CT); CT for haematoma.

Time never excludes a complication; slings can expose decades later.

4 · Midurethral sling complications

Why this matters: how much mesh to remove depends on the problem.

Vaginal exposure

- It follows infection, haematoma, excess tension or superficial placement, with bleeding, discharge, dyspareunia, hispareunia (partner pain) or UTI.
- A small early exposure gets **topical oestrogen** (1 g cream nightly for 2 weeks, then 2–3 times weekly; or estradiol 10 µg tablet).
- **Excision**: hydrodissect, incise or raise an inverted-U flap, locate the mesh (sound, ultrasound) and expose its full width. Pass an instrument behind, divide, excise at least **1 cm** around the defect and close. Avoid blind lateral dissection, which causes vulval haematoma, especially after transobturator (TO) slings.

Pain

- Risks: young age, fibromyalgia, prior pelvic pain. Causes: shrinkage, tension, obturator neuralgia, hypertonia.
- **Immediate severe pain**: exclude haematoma, consider **early removal** before ingrowth.
- Delayed focal pain gets pelvic floor physical therapy (PFPT), trigger point injection and oestrogen, then segment excision (60–90% improve). Diffuse late pain reflects central sensitisation, and excision may fail.
- Remove **as much vaginal mesh as possible** (81% pain-free). For TO **groin pain**, add a thigh incision through adductor brevis to obturator externus.

Bladder perforation

- An unrecognised trocar puncture (low body mass index, prior surgery) is avoided by cystoscopy with a **full bladder**. It causes storage LUTS, haematuria, UTI, stones or vesicovaginal fistula (VVF).
- **Endoscopic** removal (holmium, scissors, resectoscope) has low morbidity, but mesh stays in the wall and re-extrudes. **Open or robotic transvesical** removal excises 2 cm beyond the defect with two-layer closure and catheter 2–3 weeks; re-extrusion is minimal.

Urethral perforation

- Causes are trocar injury, over-tensioning and **urethral dilation** (avoid it; divide an obstructing sling). Features: voiding pain, retention, haematuria.
- Definitive repair: inverted-U flap, divide the uninjured arm, peel mesh off, minimal urethrotomy, layered closure with or without **Martius** flap, and non-overlapping suture lines. Stricture needs buccal urethroplasty.
- After excision, SUI occurs in **42%**, fewer with a simultaneous fascial sling.

Voiding dysfunction

- Midline **sling division** resolves retention in 95%, symptoms in 75%. Urgency incontinence persists in up to 76%, de novo in up to 45%. Recurrent SUI (20–74%) is treated by bulking, repeat MUS or fascial sling.
- Persistent symptoms need UDS (obstruction, underactivity or dysfunctional voiding), then wider excision and urethrolysis. A post-excision haematoma is observed unless expanding or unstable.

> **KEY POINT**
> **How much to remove**: retention needs midline division; exposure at least 1 cm each side; pain maximal vaginal removal, rarely arms. **Never remove asymptomatic mesh**; more removal means more bleeding and more POP and SUI recurrence.

5 · Transvaginal POP mesh

Why this matters: withdrawn meshes still cause problems.

- In the largest randomised trial (n = 1358), native and mesh repair were equal at 6 years (re-operation 12% vs 10%); 8% of the mesh arm needed excision.
- Kits (Perigee, Apogee) differ in arm routes, so get the operative note. Exposure is 10–12%, removal 5–8% (up to 8 years). **Contraction** (~12%) is unique to POP mesh and causes tightness and dyspareunia. Bladder perforation is 0.7–3.5%, rectal 0.15%, fistula 0.2%.
- Treat **conservatively** (observe, oestrogen) if <1 cm, mildly symptomatic and uninfected; only **size** predicts success. After office trimming, 73% later need theatre.
- **Excision**: examination under anaesthesia, cystoscopy or proctoscopy, thick flaps, then clamp behind the mesh and divide (the **most dangerous step**: most visceral injuries). Cut arms under tension, don't chase them, then repair natively.
- Afterwards: cystocele recurrence 19%, repeat excision 20%, VVF 3%; multicompartment excision raises transfusion (1% to 9%); 45% abstain sexually.

6 · Sacrocolpopexy mesh

Why this matters: infected deep mesh may need major abdominal surgery.

- Exposure is ~**10.5% at 7 years**, higher with total hysterectomy (23% vs 5%) and stage 3–4. Cervix-sparing surgery has low risk.
- Features are discharge, granulation, dyspareunia, cuff abscess, bladder or bowel perforation and **discitis**; image cross-sectionally.
- Cut an **exposed suture** below the knot in the office. Small exposures: oestrogen 6–8 weeks (low success). Vaginal excision often fails; partial colpocleisis succeeds in ~50% but shortens the vagina, so not if sexually active.
- **Abdominal removal** suits failed vaginal repair, pain, infection, discitis or ureteric complication. **Remove the entire infected mesh**, including sacral sutures, because residual mesh causes sinus, abscess and fistula. Partial vaginectomy is common (74%); bowel injury up to 29%.

7 · Male slings

Why this matters: male slings cause retention and pain more than exposure.

- Fewer complications: clean wound, deep cover, thick bulbar urethra.
- **Advance XP** causes retention in 14–21%, usually for days; incise one arm if it persists. Neobladder patients are at highest risk.
- **Virtue** causes perineal or scrotal pain in 14–17%, chronic in 0–7%. Divide an InVance before an artificial urinary sphincter (AUS).
- **Urethral extrusion** usually follows an unrecognised perforation without intraoperative cystoscopy. Do urethroscopy in persistent retention; excise lateral to the urethra, repair primarily or with buccal graft, avoiding transection.

121 Urinary Tract Fistulae

1 · Definition and universal principles

Why this matters: the same principles govern every fistula repair.

- A **urinary tract fistula** is an abnormal epithelial- or mesothelial-lined connection from the urinary tract to another cavity (bowel, vessel, genital tract) or skin.
- Causes: iatrogenic (commonest in the developed world), malignancy, inflammation, infection, ischaemia, trauma, childbirth.
- It persists because **urine keeps flowing through a tract that cannot heal**, so it epithelialises. Every principle diverts flow or restores healing.
- The **first repair has the best chance**; optimise patient and surgeon first.

Management (whole course) — Table 121.1	Operation (any route) — Table 121.2
Optimise nutrition	Evaluate concomitant injury; rule out malignancy
Control infection	Adequate exposure; remove all foreign material
Continuous drainage (catheter/stent)	Tension-free closure; vascularised edges
Eliminate obstruction	Multiple, watertight, **non-overlapping** layers
Consider malignancy — **always biopsy** if any history	Consider interposition; drain after

KEY POINT
Trap: a fistula may be the only sign of recurrent cancer, so biopsy the tract after prior malignancy or radiation.

2 · Vesicovaginal fistula (VVF) — aetiology

Why this matters: cause predicts size, site and repair timing.

VVF is the commonest acquired urinary fistula (**>75%** of genitourinary fistulae).

	Developing world — obstetric (>90%)	Developed world — iatrogenic
Cause	Obstructed labour; young, small pelvis, poor nutrition	Unrecognised surgical injury; **>75%** after hysterectomy
Mechanism	Fetal head impacted for days causes pressure necrosis and a large defect	Dissection, cautery, clamp crush, bladder caught in cuff closure
Extent	Trigone, urethra, ureter often involved; "obstructed labour injury complex" (renal, rectal, orthopaedic, neurological, psychosocial)	Solitary, **supratrigonal at vaginal cuff**

- **Hysterectomy** causes VVF in ≈ 0.8/1000: lowest vaginal, highest laparoscopic (highest 30-day rate after laparoscopically assisted vaginal hysterectomy). Distorted anatomy (endometriosis, fibroids, prior surgery or radiation, malignancy, bleeding) and concomitant prolapse, incontinence or mesh surgery raise risk. Immediate recognition and repair prevents fistula.
- **Universal cystoscopy at hysterectomy** raises intraoperative detection, has inconsistent effect on delayed injury, and adds time and urinary tract infection (UTI). AUGS (urogynaecology) advises it with any prolapse or anti-incontinence procedure, AAGL (laparoscopy) after laparoscopic or robotic hysterectomy; ACOG (obstetrics) makes no recommendation after benign hysterectomy.

- **Radiation** causes obliterative endarteritis and fibrosis, then necrosis. VVF follows 0.6–2% of cervical cancer radiotherapy, typically **1.5–2 years** later (up to 30); risk is dose-dependent, worse in smokers.
- Others: mesh (sustained inflammation), **bevacizumab** (anti-VEGF, causing hypoxia), cancer; rarely pessary, intrauterine device, infection, lasers, endometriosis.

3 · Classification of VVF

Why this matters: classification predicts closure and continence.

Of >25 systems, the WHO uses simple vs complex.

Axis	Goh	Waaldijk
Basis	Distance of distal edge from external urethral meatus	Closing (continence) mechanism and urethral involvement
Types	1: >3.5 cm · 2: 2.5–3.5 · 3: 1.5–2.5 · 4: <1.5 cm	I: closing mechanism spared · IIA: no (sub)total urethral involvement · IIB: (sub)total urethral involvement (A/B = without/with circumferential defect) · III: miscellaneous (ureteric)
Size	a <1.5 · b 1.5–3 · c >3 cm	Small <2, medium 2–3, large 4–5, extensive ≥6 cm
Tissue	i none/mild fibrosis · ii moderate/severe · iii special (radiation, ureteric, circumferential, previous repair)	—

KEY POINT

Trap: Goh type 1 is farthest from the meatus (best) and type 4 closest (worst); closer means worse closure and continence. Goh predicts closure better. Always describe size, location (trigonal vs supratrigonal), tissue loss, and ureteric and urethral involvement.

4 · Diagnosis of VVF

Why this matters: a missed ureteric injury or cancer dooms the repair.

- The classic symptom is **continuous leakage day and night**, in any position; a large tract abolishes the desire to void, while a small one leaks positionally. Recurrent UTI, haematuria or "discharge" occur.
- **Timing points to mechanism**: a missed cystotomy leaks in the first days, thermal or crush injury at **7–10 days**, and radiation after years. Fever, flank pain or ileus suggest urinoma or ureteric injury.
- On speculum, the apex is commonest after hysterectomy; vaginal fluid creatinine ≈ urine creatinine. Biopsy friable tissue after cancer or radiation, examine under anaesthesia if stenosed, give vaginal oestrogen for atrophy, and optimise smoking, glucose and nutrition.
- **Single dye test**: backfill ~300 mL methylene blue, insert a tampon, walk 15–60 min. Proximal stain means VVF; distal means urethrovaginal fistula or stress urinary incontinence (SUI); dye from the cervical os means vesicouterine fistula.
- **Double dye test**: oral phenazopyridine 100–200 mg (orange in 1 h) plus blue in the bladder, then a tampon. **Orange means ureterovaginal fistula (UVF)**, blue means VVF, both means both.
- **Cystoscopy is integral** (sensitivity ~92%): it shows size, number, orifices, bladder neck, mesh and capacity, and allows biopsy; probe small tracts with a guidewire or 5-Fr catheter.

Under anaesthesia, add retrograde pyelography and stenting. **Vaginoscopy** shows the vaginal side, allows biopsy and overcomes radiation stenosis.

- Cystogram or voiding cystourethrogram (VCUG) shows vaginal filling (best on lateral view) and any urethrovaginal fistula. T2 MRI helps difficult cases.
- **Upper tract imaging is mandatory**, because ureteric injury coexists in up to **12%**. Use CT urography (CTU) with a delayed phase, or MR urography if contrast allergy or renal failure. **Partial ureteric obstruction with VVF means UVF until proven otherwise.** If unclear, do retrograde pyelogram and stent, or percutaneous nephrostomy (PCN) if that fails.
- Urodynamics (UDS or video-UDS) helps planning: 10–15% have SUI (autologous sling), and radiation may reduce capacity and compliance (augmentation or diversion). Occlude a large fistula with a Foley balloon or pack to fill.

> **KEY POINT**
> Work-up order: is it urine (creatinine), where is it from (dye tests, endoscopy), and what else is damaged (hence compulsory upper tract imaging). A preoperative PCN also allows antegrade ureteroscopy to find the ureter in scar.

5 · VVF — timing, conservative care, repair

Why this matters: timing and route decide the first repair.

- P3 P2 P1 VVF management options; early vs delayed repair; eponymous techniques — *catheter trial (~15%, <1 cm, within 3 weeks, EAU ≥12 weeks), fulguration, sealants not recommended; timing by cause; vaginal vs abdominal route (~90% either); flap-splitting, Latzko, cuff excision, O'Conor, transvesical/extravesical, lap/robotic; flaps; postoperative care; diversion; outcomes*

Not in the paper answer:

- Size ≈ >3 cm predicts failure (≥1.5 cm for residual obstetric incontinence).
- Apical VVF (8–10 cm from introitus) lies beyond Martius reach; use peritoneum or omentum.
- Radiation strongly predicts failure and brings ureteric stricture, poor compliance and intrinsic sphincter deficiency (ISD); consider diversion.
- O'Conor fails with carcinoma (with or without radiation), wound infection or poor mobilisation.
- Robotic O'Conor closes the vagina **transversely** and the bladder **vertically**, so the lines cross.

Setting	Timing of repair	Rationale
Obstetric (traditional)	≥3 months, catheter meanwhile	Inflammation settles, necrosis sloughs
Iatrogenic, clean tissue	Early; European Association of Urology (EAU): no minimum delay if no infection or recent radiation	Comparable closure; less distress
Radiation-induced	≥ **12 months** after radiation ends	Late injury still evolving

- Success has no consensus definition; anatomical closure is not continence.
- **Size**: tracts <3 mm close spontaneously best, and <10 mm may close with drainage.
- **Catheter** drainage is favourable for tracts <1 cm catheterised within **3 weeks**; it is unfavourable at ≥6 weeks or with radiation or malignancy (never heal). The EAU trial lasts ≥**12 weeks**. Sealants are possible for <1 cm in poor surgical candidates (risks: catheter adherence, spasm, stone on glue).
- **Location**: within **1 cm** of a ureter, stent it; if the ureter is involved, reimplant (abdominal). Bladder neck involvement needs reconstruction, plus a pubovaginal sling if needed.

- **Route**: there are no RCTs; first-attempt closure is ≈ **90%** either way. Transvaginal repair has less blood loss, pain and stay; some call it gold standard.
- Abdominal suits large fistulae (≥4 cm primary, ≥2 cm recurrent), urethral, bladder-neck or ureteric involvement, a non-capacious vagina, radiation, or no lithotomy. Vaginal suits dense adhesions or avoiding general anaesthesia, but risks shortening and dyspareunia (especially Latzko); abdominal spares sexual function.
- Tract excision and interposition are unproven in simple VVF but valued in irradiated, recurrent or poor tissue.

ALGORITHM

Transvaginal flap-splitting (Table 121.5)

1. Cystoscopy: stent close ureters; wire through the tract.
2. Suprapubic catheter (SPC) and urethral catheter; episiotomy if narrow.
3. Foley ≥8 Fr through the fistula for traction; hydrodissect.
4. **J- or U-incision** with a 1–3 mm cuff; raise wide anterior and posterior flaps.
5. Layer 1: 3-0/4-0 delayed absorbable. Layer 2: **perpendicular** to layer 1.
6. Leak test 200–300 mL; Martius or peritoneal flap if needed.
7. Advance the opposite flap ≥2 cm beyond (layer 3); cystoscopy for efflux.

- **Postoperative drainage is the most important part**: urethral catheter, often with SPC, for ≥10–14 days (two guard against blockage); cystogram is optional. Give anticholinergic or β3-agonist for spasms, check drain creatinine for urinoma, and remove stents at 4–6 weeks. †Advise pelvic rest for ~6 weeks.
- **Latzko** leaves the bladder unopened and the tract unexcised: denude 1–2 cm and close in 2 or more superficial layers. It is short, avoids ureteric dissection and succeeds in ≥90% non-irradiated; it is favoured after radiation or with close orifices, but shortens the vagina.

ALGORITHM

O'Conor (Table 121.7)

1. Ureteric catheters; low midline extraperitoneal approach.
2. Sagittal dome cystotomy; **bivalve** down around the fistula; excise the tract.
3. Separate bladder from vagina ≥2–3 cm beyond; close the vagina.
4. Place **omentum or peritoneum** beyond the vaginal suture line.
5. Close bladder in 1–2 layers; SPC with or without urethral catheter; fill test.
6. Drain 10–14 days, then cystogram.

- O'Conor causes more de novo urgency incontinence (UUI) and lower urinary tract symptoms (LUTS) from long suture lines. Transvesical repair (limited cystotomy) restricts mobilisation and interposition; it suits low-lying fistulae. Extravesical repair hides the orifices (ureteric risk) and has a scarred plane; it suits supratrigonal, not infratrigonal, fistulae.
- **Robotic O'Conor** uses an omental J-flap, epiploic appendage or peritoneum; indocyanine green (ICG) checks edge vascularity. Dense adhesions contraindicate it.

6 · Interposition flaps

Why this matters: flaps help poor tissue, not simple fistulae.

- Flaps reinforce, fill dead space and import blood and lymphatics. Use them for **complex VVF** (radiation, recurrent, ischaemic obstetric, poor tissue); an RCT showed **no benefit of**

Martius in simple VVF.

- **Martius** supply: internal pudendal (inferior), external pudendal (superior), obturator (lateral). Through a 4–7 cm labial incision, mobilise lateral to bulbocavernosus, keep the **superior** pedicle for VVF (inferior for rectovaginal), tunnel ~2 cm, and drain 48 h. It reaches mid-distal VVF, bladder neck and urethra, but **not the apex**.
- A vaginal peritoneal flap (pouch of Douglas) suits the apex; **omentum** (gastroepiploic) suits radiation, recurrent or large fistulae (if short, use abdominal peritoneum or sigmoid epiploic appendages). Rectus, gracilis and Singapore flaps are a last resort.
- After repair, residual UUI or SUI (predicted by Goh 3–4, >1 repair, positive cough or pad test) drives quality of life and depression; the International Continence Society (ICS) advises physiotherapy for all obstetric fistula. A small vagina and fistula ≥3 cm predict dyspareunia. Divert if unrepairable (resource-poor: ureterosigmoidostomy).

7 · Ureterovaginal fistula (UVF)

Why this matters: early stenting often avoids an operation altogether.

- P3 Urological complications in OBS-GYN and uretero-vaginal fistula — *distal ureteric injury at hysterectomy, danger sites, normal voiding + leak within 4 weeks, double dye/CTU/RGP, stent within 2 weeks for 4–6 weeks, reimplant ± psoas hitch/Boari, ≥90%, follow 30 months*

Not in the paper answer:

- Preoperative intravenous pyelography does not help; prophylactic stents are selective (e.g. caesarean hysterectomy for placenta accreta).
- Up to **25%** of VVF have a coexisting UVF; delayed recognition raises fistula risk ~4-fold. Ileal ureter or nephropexy adds length; early and ≥6-week repair succeed equally.

ALGORITHM

1. Suspect UVF; exclude VVF; confirm with CTU or retrograde pyelogram.
2. Stent; remove at 4–6 weeks and reimage.
3. If resolved, stop. If persistent or stenting fails, then do ureteroneocystostomy.

KEY POINT

UVF: wet but voids normally. The ureter heals over a stent if reached early, so the detection-to-stent interval largely decides whether surgery is needed.

8 · Vesicouterine fistula (VUF)

Why this matters: VUF presents with cyclical haematuria, not leakage.

- It follows **low transverse caesarean** and joins the uterine body above the internal os to the posterior bladder; a competent cervix **preserves continence**.
- **Youssef syndrome**: menouria, amenorrhoea, first-trimester miscarriage, continence. Incontinence occurs only if the tract is below the os or the cervix incompetent. Bladder endometriosis is the differential of menouria.
- Dye appears from the cervical os; cystoscopy shows a posterior supratrigonal defect (the scope may enter the uterus). Add cystogram, hysterosalpingography or MRI, and image the upper tract.
- Spontaneous closure is ~5%. After 3–4 months (involution), separate transabdominally and close in layers with omentum or peritoneum (uterus-sparing), or do hysterectomy if the family

is complete; Martius rarely reaches. Then use contraception until healed and deliver by caesarean.

9 · Urethrovaginal fistula

Why this matters: closure is easy but continence is hard, so counsel accordingly.

- P3 Urological complications of OBGY surgery and urethrovaginal fistula — *diverticulectomy commonest Western cause (20–25%), distal vs proximal symptoms, flexible urethroscopy, ~20% separate VVF, flap repair + Martius, autologous sling never synthetic, SPC ≥14 days, closure 90–95% vs continence 50–80%*
- Mesh may cause hispareunia (partner pain).

10 · Rectourethral fistula (RUF)

Why this matters: tissue quality, not size, decides the operation.

- P3 Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management — *post-prostate-cancer treatment, faecaluria/pneumaturia, biopsy, colostomy first, transperineal gracilis vs York-Mason vs abdominal, salvage diversion*

Martini	Category
Stage (size)	I ≤1.5 cm · II >1.5 cm · III any size with sphincter damage
Grade (tissue)	0 surgical injury only · 1 prior radiation/energy · 2 salvage radical prostatectomy (RP) or salvage ablation

11 · Vesicoenteric fistula (VEF)

Why this matters: the cause lies in the bowel.

- Colovesical is commonest. M : F is **3 : 1** because the uterus shields the bladder; affected women often had a hysterectomy.
- Causes in order: **diverticulitis**, **Crohn disease** (ileovesical), colorectal cancer (10–15% of colovesical fistulae have a neoplasm), then bladder or prostate cancer, radiation, appendicitis. †The sigmoid typically joins the left posterolateral bladder or dome.
- It causes **pneumaturia** (50–70%), faecaluria and polymicrobial UTI. **Gouverneur syndrome** is suprapubic pain, frequency, dysuria and tenesmus.

Test	Role
Poppy seed test	Highest detection of presence; cheap; no location
CT (preferred)	Air in bladder, wall thickening, adherent bowel; shows cause
Cystoscopy	Nonspecific erythema/oedema
Colonoscopy	Poor for tract; **excludes cancer**
MRI	Crohn, anatomy

- Treat conservatively if symptoms are minimal and disease benign (especially Crohn: immunomodulators, nutrition); spontaneous closure is <2%.
- **Resect the diseased bowel with the fistula**, with single-stage anastomosis (plus loop ileostomy if needed), increasingly laparoscopically; use Hartmann for abscess, contamination, advanced cancer or radiation. Leave or oversew a small bladder defect, do **partial cystectomy** if malignant, and interpose omentum.

> **KEY POINT**
> VEF is a bowel disease presenting in the bladder: remove the diseased bowel, and the bladder usually heals itself.

Renocolic fistula: spontaneous ones are mostly renal (xanthogranulomatous pyelonephritis [XGP], chronic infection) and need nephrectomy, plus bowel resection if needed. Small iatrogenic ones (colonic injury at percutaneous nephrolithotomy [PCNL]) settle with antibiotics, stent or nephrostomy, and nil by mouth.

12 · Urocutaneous fistulae

Why this matters: downstream obstruction keeps these open.

- **Vesicocutaneous** fistula follows long-term SPC, open prostatectomy or radiation, and persists with an epithelialised tract, infection, foreign body, tumour or bladder outlet obstruction (**BOO**). Drain, catheterise and relieve BOO; if unhealed, do fistulectomy or partial cystectomy.
- **Urethrocutaneous** fistula is the commonest hypospadias complication (up to 20%). Wait ≥ **6 months** and exclude distal stricture. Close <1 cm locally; ≥1 cm or redo needs vascularised flaps (**tunica vaginalis** better than dartos). Massive loss needs gracilis; neurogenic patients with pressure ulcers need an ileal conduit.
- **Nephrocutaneous** fistula follows stones, XGP, prior surgery or tuberculosis (≥10%). The kidney is non-functioning in ~90%, so do **nephrectomy**; use contrast CT.

13 · Pubovesical fistula (PVF)

Why this matters: it is really pubic osteomyelitis.

- A tract from bladder, prostate or urethra to the pubic symphysis causes **osteomyelitis**. It follows prostate radiotherapy plus repeated outlet procedures (transurethral resection, bladder neck incision, stricture surgery), median ~10 years after radiation, and also artificial urinary sphincter (AUS) surgery.
- It presents with **pubic pain on walking**, abscess, urosepsis, sinus. **MRI is most sensitive and specific**; CT-guided bone biopsy grows Candida, Enterococcus or Pseudomonas (urine culture is unreliable).
- Conservative care usually fails. Debride bone, excise the fistula, divert, fill with rectus or omental flap, and give 4–6 weeks of antibiotics. Repair primarily only with a working sphincter, normal bladder and no distal obstruction; otherwise, **cystoprostatectomy and diversion**.

14 · Urovascular fistulae

Why this matters: these can bleed catastrophically.

- **Arteriocalyceal** fistula follows PCNL, transplant or trauma, sometimes with pseudoaneurysm or arteriovenous fistula; treat by **angioembolisation**.
- **Ureteroarterial fistula (UAF)** is an emergency. Most are secondary (85%): long-term stents, pelvic surgery, radiation, cancer, vascular grafts; the common iliac is commonest. It presents with **intermittent haematuria**, and ~20% are unstable.
- Diagnose with **angiography** (provoke by stent removal only with endovascular treatment ready); cystoscopy shows pulsatile bleeding from one orifice. CT angiography is insensitive; a

negative test does not exclude it.

- Treat with an **endovascular stent graft** with or without internal iliac embolisation; use open surgery (bypass, ligation, nephrectomy) for infection, abscess or enteric involvement.

15 · Renal urine leak and anastomotic leak

Why this matters: most leaks settle with drainage.

- After **trauma**, a leak occurs in 1–7% (~a quarter of grade IV–V). Use CT with a 10-min delayed phase; a non-opacified distal ureter suggests ureteropelvic junction avulsion, needing retrograde pyelogram. Most resolve; persistent leaks need stent or nephrostomy.
- After **partial nephrectomy** (0.5–20%; low after robotic), a central tumour predicts leak. Check drain creatinine and CT; drain, adding stent or nephrostomy if needed; refractory leaks need glue or embolisation.
- **Vesicourethral anastomotic leak after RP** (Patil): I extraperitoneal <6 cm, II extraperitoneal ≥6 cm, III intraperitoneal. Prolong the catheter, then withdraw the drain, then place stents or nephrostomies; reoperate last.

16 · Fistula after gender-affirming surgery

Why this matters: fistula is common after phalloplasty.

- After transmasculine surgery (phalloplasty, metoidioplasty), urethrocutaneous fistula and stricture are common (~3 months); **primary colpectomy** roughly halves fistula. Small ones close with a catheter; persistent ones need excision or regrafting.
- After transfeminine surgery, urethro- or vesiconeovaginal fistula is rare; divert suprapubically, then repair with interposition.

17 · Landmark trials

Why this matters: these trials guide flap and stent use.

Trial	Design	Result	Take-home
Martius flap in simple obstetric VVF	RCT, repair ± Martius	No difference in closure; ~⅓ donor-site pain/numbness	Reserve Martius for complex fistula
Prophylactic ureteric stents in gynaecological surgery	RCT (+ meta-analysis)	No/small reduction in injury (number needed to treat 224)	Selective, not routine, stenting

122 Bladder and Female Urethral Diverticula

1 · Bladder diverticula — definition and pathology

Why this matters: a sac without muscle cannot empty, causing every complication.

- P2 Bladder diverticulum. — *definition, congenital vs acquired, investigations, management, diverticulectomy steps, T3a staging*

Not in the paper answer:

- Diverticular cancer: radical cystectomy gives the same stage-for-stage survival; upstaging is common.

- A diverticulum is urothelium herniating through the muscularis propria. Its wall has mucosa, lamina propria, scattered non-functional muscle and adventitia; an outer pseudocapsule gives the surgical plane.
- With **no functional muscle, the sac cannot contract**, so post-void residual (PVR) is high and stasis leads to infection, stones and malignancy. Middle-detrusor fibres ring the neck, giving a narrow ostium (target of endoscopic resection).
- About 90% occur in adults; M : F ≈ **9 : 1**. Cellules, saccules and diverticula form a pressure continuum.

Feature	Congenital	Acquired
Age / number	<10 years; solitary	Men >60; often multiple
Site	Posterolateral to ureteric orifice (~90% near ureterovesical junction [UVJ]); vesicoureteral reflux (VUR) common	Ureterovesical hiatus commonest; anywhere
Cause	Detrusor weakness at UVJ	Bladder outlet obstruction (**BOO**) or neurogenic — almost always
Wall / presentation	Smooth; acute urinary tract infection (UTI)	Trabeculated; incidental (lower urinary tract symptoms [LUTS], haematuria, UTI)
Malignancy	Not increased	Increased

- Up to 60% of congenital cases have a syndrome, neuropathic bladder or BOO, e.g. **Menkes** (copper, kinky hair), **Williams**, **Ehlers-Danlos**, fetal alcohol, prune-belly, posterior urethral valves. A single simple lesion needs no chromosomal testing.
- Acquired cases are due to benign prostatic hyperplasia in ~**70%**, also bladder neck obstruction and stricture; in women, dysfunctional voiding, prolapse, primary bladder neck obstruction and post-sling obstruction.
- A **Hutch diverticulum** is paraureteric, in a neuropathic bladder with VUR (paraplegics).

> **KEY POINT**
> Congenital: weak wall, normal pressure. Acquired: normal wall, high pressure, so an acquired diverticulum sends the search back to the outlet.

2 · Bladder diverticula — evaluation

Why this matters: work-up must find the cause and exclude cancer.

- Symptoms are nonspecific (incomplete emptying, double voiding); rarely the sac compresses the iliac vein or lies in an inguinal hernia. Persistent pyuria despite antibiotics may justify

excision.

- Do rectal examination, PSA, urinalysis, culture, **cytology**, endoscopy and upper and lower tract imaging; seek BOO or occult neuropathy.
- Voiding cystourethrogram (**VCUG**; AP, oblique, lateral) shows size, VUR (up to 13% congenital) and emptying. **Paradoxic enlargement**: the sac fills on voiding and refills the bladder afterwards.
- CT or MRI shows number, masses and relation to ureters and rectum; use it if the neck is obstructed.
- Hydroureteronephrosis (HUN) occurs in ~7% of adults (upper tract abnormality in up to 30% of children), from compression, underlying dysfunction or VUR. The pelvic ureter deviates **medially**; a ureter entering a sac loses its backing and refluxes.
- Video-urodynamics (**VUDS**) shows a **pressure-sink effect**: contractility looks falsely low, and a raised PVR may be diverticular urine.
- **Cystoscopy** inspects the whole sac (flexible scope) with cytology from it. Biopsy with extreme care, as perforation seeds tumour. CT can miss tumour.
- With VUR and a congenital diverticulum, reimplant **selectively** (scarring, deteriorating function). An untreated cause brings recurrence and voiding problems; treating it may make excision unnecessary.

> **KEY POINT**
> Two jobs: find the cause (outlet or neurogenic) and exclude cancer in a sac each test sees imperfectly, so tests are combined.

3 · Malignancy in bladder diverticula

Why this matters: missing muscle lets tumours spread early.

- Stasis and inflammation drive it; prevalence 0.8–10%, peak 65–75 years. Urothelial carcinoma is **70–80%** and squamous cell carcinoma (SCC) 20–25%.
- With **no muscularis propria**, spread is early and extravesical, transurethral resection (TUR) risks perforation and seeding, and staging is inaccurate. Clinical stage is the key prognostic factor.
- **Radical cystectomy** gives the same stage-for-stage survival. Partial cystectomy or diverticulectomy suits selected low-grade, low-stage tumours; some favour open excision without prior TUR. TUR with intravesical therapy has an undefined role.

> **KEY POINT**
> Treat a diverticular tumour as potentially invasive from the outset; missing muscle explains early spread, perforation risk and inaccurate staging.

4 · Bladder diverticula — management

Why this matters: treat the outlet first; operate for complications, not size.

ALGORITHM

1. Assess with history, cystoscopy, cytology and VCUG.
2. If a lesion is suspicious, then biopsy and give cancer therapy.
3. If not, image the upper tract. If HUN or symptoms, then do urodynamics and treat BOO, then reassess.
4. If a problem from the diverticulum persists: fit → diverticulectomy; unfit → clean intermittent catheterisation (CIC).
5. If asymptomatic and uncomplicated, then survey.

- **Diverticulectomy indications**: refractory LUTS, relapsing UTI, **stones**, **carcinoma or premalignancy**, upper tract deterioration (obstruction, VUR). **Size alone is not an indication.**
- **Treat the outlet first or simultaneously**; a small sac may then empty (transurethral resection of the prostate [TURP], reassess, then surveillance). Concomitant prostatectomy is common.
- If observing, counsel about cancer and survey with symptoms, cytology and cystoscopy (no fixed schedule; natural history unknown). The unfit with poor emptying need CIC or an indwelling catheter.
- **Endoscopic** treatment (unfit, or at TURP) resects the neck to muscle, sometimes fulgurating the lining; "venting" risks retention, and large sacs do poorly.
- Diverticulectomy is open, laparoscopic or robotic *(Ch 124)*; †use an ipsilateral ureteric catheter, dissect intra- or extravesically in the pseudocapsule, close the neck in layers, and reimplant a ureter entering the sac.
- Observe syndromic multiple diverticula in children (poor healing, recurrence).

5 · Female urethral diverticulum (UD) — anatomy and aetiology

Why this matters: anatomy explains why repair risks continence.

- P1 Female urethral diverticula — *Routh theory, posterolateral ostium, three Ds ~5%, stones 4–10%, adenocarcinoma, imaging sensitivities (MRI 97%), L/N/S/C3, diverticulectomy principles, autologous sling only, recurrence 10–22%, de novo SUI 7–16%*

Not in the paper answer:

- Cancer in UD recurs after local excision alone.
- De novo SUI risks (≥30 mm, proximal, circumferential); ~50% of preoperative SUI improves.
- The urethra is 3–4 cm; the urethropelvic ligament has an abdominal leaf (endopelvic fascia) and a vaginal leaf (**periurethral fascia**); UD lie between the leaves.
- **Periurethral glands** lie posterolaterally in the distal two-thirds, with ducts draining into the **distal third**. Skene glands are largest, most distal and lateral to the meatus.
- The striated sphincter is a mid-urethral horseshoe, deficient vaginally; a UD overlaps it, so plan for continence. Proximal urethra: urothelium, iliac nodes; distal: squamous, inguinal nodes.
- **Young–Wahle** sequence: a gland abscess disrupts muscle, herniates into the fascia (usually **ventrally**), then ruptures into the lumen. Routh implicated E. coli and gonococcus.
- Against the gland theory, ~⅓ are proximal. Bulking agents mimic UD; mesh slings do not cause it. UD is found in proportion to how hard it is sought.

- Configuration: simple ~73%, saddlebag ~17%, circumferential ~10%. Site: mid-urethral ~50%, proximal or distal ~25% each; ~6–19% multiple.
- Carcinoma in UD is **adenocarcinoma** > urothelial > SCC (primary female urethral cancer is mostly SCC). Imaging misses small cancers, so biopsy if treatment is non-excisional. Local excision alone recurs: use anterior exenteration or cystourethrectomy with or without radiotherapy (RT), or excision plus RT or chemotherapy. Nephrogenic adenoma mimics adenocarcinoma.

6 · UD — presentation and evaluation

Why this matters: UD masquerade as common LUTS for years.

- All three Ds occur in only ~**5%**. Features: irritative LUTS, pain, recurrent UTI, mass (~⅓), incontinence (~⅓), haematuria, discharge, retention; 10–20% are asymptomatic. **Size does not track symptoms**, and UD is misdiagnosed for years (interstitial cystitis, vulvodynia), so examine.
- In pregnancy, give antibiotics or aspirate; repair after delivery.
- A mass lies 1–3 cm inside the introitus; meatal distortion suggests Skene, and a **hard mass suggests stone or cancer**. Check atrophy (oestrogen), stress test, prolapse. A sterile culture does not exclude infection.
- **Cystourethroscopy** shows the ostium at **4 and 8 o'clock** mid-urethra in 15–89%; not seeing it must not stop work-up.
- **VUDS** shows SUI in up to 50% and separates true SUI from paradoxic dribbling from the sac; the pressure profile is biphasic.
- **MRI is best** (T2 high, no voiding needed, maps extent; may miss small cancers, costly). Double-balloon positive-pressure urethrography (Trattner) is rarely available; ultrasound shows an anechoic sac; VCUG needs a voiding or postvoid film and underestimates size. Bulking agent on MRI: multiple, upper urethra, no fluid level or ostium, lower T2 than urine.

> **KEY POINT**
> UD symptoms mimic common LUTS and do not track size, so suspicion and examination make the diagnosis. Imaging confirms and **maps** it, especially extension towards the bladder neck (bladder or ureteric injury, postoperative incontinence).

7 · Differential diagnosis of periurethral masses

Why this matters: only UD communicates with the urethra.

Lesion	Clue	Treatment
UD	Mid/proximal; communicates	Excision, reconstruction
Leiomyoma	Firm, mobile, solid; oestrogen-dependent	Transvaginal enucleation
Skene cyst/abscess	Lateral to and distorts meatus; non-communicating	Aspiration, marsupialisation, excision
Gartner duct cyst	Anterolateral; may receive ectopic ureter — image upper tract	Observe, sclerotherapy, excision
Vaginal wall cyst	Müllerian commonest	Excise if symptomatic
Urethral prolapse	Red "doughnut"; prepubertal, postmenopausal	Oestrogen; excision
Caruncle	Red, †posterior meatal lip; postmenopausal	Oestrogen; excise if atypical

Lesion	Clue	Treatment
Bulking agent	Injection history	Excise if symptomatic

Skene gland adenocarcinoma may raise PSA.

> **KEY POINT**
> Two questions sort most periurethral masses: does it communicate (only UD), and where is it relative to the meatus (Skene, prolapse and caruncle lie at the meatus)? A firm, mobile, solid mass is a leiomyoma.

8 · Classification and SUI

Why this matters: coexisting SUI needs surgery only if bothersome.

System	Basis	Use
L/N/S/C3 (Leach)	Location, Number, Size, Configuration, Communication, Continence	Standard; not validated
Ginsburg–Genadry	Location	Distal: marsupialisation; proximal: excision
Leng–McGuire	Periurethral fascia present?	Absent = **pseudodiverticulum**, needing flap/graft

- Significant de novo stress urinary incontinence (SUI) occurs in **7–16%**; risks are size ≥30 mm, proximal or circumferential sac, and short urethra. Most settles within 12 months.
- About 50% of preoperative SUI improves after excision; operate only for **bothersome** SUI: an **autologous pubovaginal sling, never synthetic**.

9 · Surgical repair of UD

Why this matters: complete excision and layered closure prevent recurrence.

- Operate for symptoms, stones or suspected cancer; treat the asymptomatic or unfit with suppressive antibiotics and postvoid stripping.
- **Spence-Duckett marsupialisation** suits only distal-third UD in women not sexually active (risks SUI, vaginal voiding, dyspareunia). Unroofing, fulguration and oxidised cellulose are abandoned; laparoscopic, robotic or suprameatal routes for anterior or proximal UD are reported.
- Principles (Box 122.2): vascularised flap, separate fascia, excise ostium and entire sac, watertight multilayer non-overlapping closure, close dead space, and **preserve or create continence**.

ALGORITHM

1. Lithotomy and Foley; suprapubic catheter (**SPC) at the start**, never the end (it would cross the repair).
2. Make an **inverted-U incision** and raise the vaginal flap off the fascia.
3. Incise the fascia **transversely** and raise fascial flaps.
4. Excise the entire sac to the ostium, including the inflamed urethral edge.
5. Close the urethra watertight over a catheter (as small as 12 Fr).
6. Close fascia **perpendicular** to it, obliterating dead space.
7. If tissue is poor or fascia attenuated, then add a **Martius** flap.
8. Advance the vaginal flap (3 layers; 4 with Martius); antibiotic pack.

- A pseudodiverticulum needs Martius or fascial graft; a circumferential UD may need segmental urethral resection.
- Give antibiotics 24 h and antispasmodics; do a **pericatheter VCUG at 14–21 days**, repeated weekly if leaking.

10 · Complications

Why this matters: recurrence, fistula and incontinence are the three failures.

Complication	Incidence	Notes
Recurrent UD	**10–22%** (pooled ~10%)	Incomplete excision, dead space; circumferential, complex, proximal, multiple
Incontinence	1.7–16%	Conservative 12 months, then autologous sling
Urethrovaginal fistula	0.9–8.3%	Distal: may need no repair; proximal: repair ± Martius
Stricture	0–5%	—
Recurrent UTI	up to 31%	Investigate for recurrence
Others	—	Distal urethral necrosis, bladder/ureteric injury, dyspareunia

Redo excision gives less relief and more fistula. Image persistent symptoms for recurrence or stricture.

KEY POINT

Complete sac removal, watertight closure, off-axis fascia and obliterated dead space prevent the three failures: recurrence, fistula, incontinence.

Feature	Bladder diverticulum	Female UD
Origin	Herniation through detrusor	Infected periurethral gland ruptured into lumen
Cause to treat	BOO / neurogenic	Infection; coexisting SUI
Commonest cancer	Urothelial (then SCC)	Adenocarcinoma
Key study	Cystoscopy + cytology; VCUG	MRI
Operate for	Complications, not size	Symptoms or complications
Operation	Diverticulectomy + outlet treatment	Transvaginal excision, layered reconstruction

77 Sexual Function and Dysfunction in the Female

- P3 Female sexual dysfunction — *definition, prevalence, response models, DSM-5 vs ICSM/ISSWSH, causes, PLISSIT/FSFI, treatment overview by domain*

Not in the paper answer:

- Only ~8–12% have a distressing disorder; ICSM 2016 adds orgasmic illness syndrome.
- DSDS is the validated desire screen; PISQ-12/IR is the only validated prolapse/incontinence tool; random testosterone is useless.
- Flibanserin: ~half respond by 8 weeks; REMS. Bremelanotide: ≤8 doses/month; hyperpigmentation. Testosterone: stop at 6 months if no benefit.
- Surgery: genital-sparing cystectomy (sex at ~6 vs 14 weeks); combined anterior and posterior repair increases dyspareunia; TOT spares the dorsal clitoral nerve.

1 · Functional anatomy

Why this matters: urethra, clitoris and vagina work as one unit, so urethral or vaginal surgery can change sexual function.

The **distal vagina, clitoris and urethra** share blood supply and nerves and respond as one unit, the **clitoral complex**.

- The **clitoris** is the penile homologue, **separate from the urethra**. Its two corpora (incomplete pectiniform septum) continue as crura on the ischiopubic rami.
- The glans is very sensitive but **not erectile**. The dorsal nerves lie **superficially on the dorsum**, outside the tunica, divided by the deep suspensory ligament, so hood or clitoral surgery can injure them.
- **Erectile** tissue: clitoral body, crura, "vestibular" bulbs. **Non-erectile vascular** tissue (labia minora, glans, periurethral tissue, vaginal walls) has raised flow but no engorgement. Erectile tissue surrounds the urethra except vaginally.
- **Skene** glands are the prostate homologue (prostate-specific antigen positive); **Bartholin** glands are the bulbourethral homologue. MCQ The G-spot has no agreed anatomical basis.
- Labium majus: anterior third ilioinguinal (L1); posterior two-thirds perineal (S3); lateral part posterior cutaneous nerve of thigh (S2). The glans drains **directly to deep inguinal nodes**. The deep dorsal vein does not run with the arteries.

2 · Physiology of the sexual response

Why this matters: which nerve carries the trigger and which the response explains surgical damage and how drugs work.

> **ALGORITHM**
>
> 1. Touch on the clitoral complex travels in pudendal (somatic) afferents to **S2–S4**.
> 2. Parasympathetic efferents (nervi erigentes, then cavernous nerves) dilate arterioles, causing engorgement and lubrication.

The reflex has **somatic afferents and visceral efferents** and can be voluntarily augmented or suppressed. **Orgasm** is cerebral: **sympathetic** output (T12–L1, hypogastric) contracts the uterus and pudendal reflexes contract pelvic striated muscle. Resolution is sympathetic vasoconstriction.

KEY POINT

Pudendal injury removes the **trigger**; pelvic plexus or cavernous nerve injury removes the **response**.

ALGORITHM

1. Neuronal nitric oxide synthase (nNOS) makes nitric oxide (NO), which activates guanylate cyclase to make cyclic GMP (cGMP).
2. cGMP lowers Ca^{2+}, so smooth muscle relaxes and tissue engorges.
3. Phosphodiesterase type 5 (PDE5) degrades cGMP; arginase consumes NOS substrate; vasoactive intestinal polypeptide (VIP) works with NO.

Desire balances **excitation** (dopamine, noradrenaline, melanocortins, oxytocin, testosterone, oestrogen) against **inhibition** (serotonin, opioids, prolactin, progesterone). Licensed drugs for hypoactive sexual desire disorder (HSDD) act on this balance.

	Oestrogen	Testosterone
Role	Maintains **substrate**: epithelium, collagen, lubrication, acidic pH, NOS/VIP	Drives **desire**; if bound to sex hormone–binding globulin (SHBG) it is inactive
Deficiency	Thin, dry, fibrotic vagina, causing dyspareunia; risk rises at oestradiol **≤50 pg/mL**	Lower libido, arousal, sensation, orgasm
Treatment risk	—	Virilisation: early reversible (acne, hirsutism); late irreversible (voice, alopecia, clitoromegaly)

Model	Stages	Key idea
Masters & Johnson (1966), linear	Excitement, plateau, orgasm, resolution	**No desire phase**
Kaplan (1977), linear	Desire, arousal, orgasm	Adds desire
Whipple & Brash-McGreer, circular	Seduction, sensations, surrender, reflection	Reflection feeds next seduction
Basson (2002), circular	Neutrality, receptivity, arousal, **responsive desire**, satisfaction	Spontaneous desire **not needed**; intimacy is the motive

KEY POINT

Under Basson, lack of *spontaneous* desire (common after menopause) is not a dysfunction, which is why DSM-5 merged desire and arousal.

3 · Classification

Why this matters: a problem is a disorder only if it persists and distresses her.

DSM-5 (Diagnostic and Statistical Manual) requires **≥6 months** and **clinically significant distress**; low desire without distress is no disorder.

Domain	DSM-IV-TR	DSM-5
Desire and arousal	HSDD; female sexual arousal disorder (FSAD); sexual aversion disorder	**Female sexual interest/arousal disorder (FSIAD)**: **≥3 of 6**: less interest; fewer fantasies; less initiation/receptivity; less pleasure, response to cues, sensation (in 75–100% of encounters). Aversion dropped

Domain	DSM-IV-TR	DSM-5
Orgasm	Female orgasmic disorder (FOD) after normal excitement	FOD: delay, infrequency, absence **or lower intensity** in **75–100%**; normal-excitement requirement removed
Pain	Dyspareunia; vaginismus	**Genito-pelvic pain/penetration disorder (GPPPD)**, ≥1 of: penetration difficulty; pain; fear of pain; pelvic floor muscle (PFM) tensing

- **ICSM 2016** (International Consultation on Sexual Medicine) keeps HSDD and FSAD **separate**, adding **persistent genital arousal disorder (PGAD)** and orgasmic illness syndrome (malaise after sex).
- **ISSWSH** (International Society for the Study of Women's Sexual Health) splits FSAD into cognitive and **genital arousal disorder** (urological); it is not diagnosed if stimulation does not match her preferences.
- About **40%** have a lifetime sexual problem; ~**8–12%** a distressing disorder. **HSDD is commonest**, FOD second.

4 · Causes (biopsychosocial)

Why this matters: most female sexual dysfunction (FSD) has a treatable contributor.

- **Biological**: diabetes mellitus (DM), hypertension, coronary disease, thyroid disease, hyperprolactinaemia, renal failure, spinal cord injury (SCI), multiple sclerosis (MS), vulval dermatoses, incontinence, pregnancy, breastfeeding, menopause.
- **Psychological**: depression (bidirectional, so screen every HSDD), anxiety, body image, abuse; restrictive upbringing; conflict, and the **partner's erectile dysfunction (ED) or premature ejaculation (PE)**.
- **Drugs**: SSRIs, SNRIs, antipsychotics (which raise prolactin); β-blockers, ACE inhibitors, calcium channel blockers, thiazides, spironolactone; antiandrogens, GnRH agonists; opiates, alcohol.
- MCQ Combined hormonal contraceptives lower ovarian testosterone and raise SHBG; oral oestrogens, tamoxifen and thyroxine also raise SHBG.
- **Genitourinary syndrome of menopause (GSM)** causes dryness, burning, dyspareunia (often **vestibular**), urgency, dysuria and recurrent UTI. Dryness causes pain, which lowers desire, so **treat the tissue first**.
- **Cancer treatment**: castration, anti-oestrogens (†aromatase inhibitors cause severe GSM), pelvic radiotherapy (stenosis), nerve injury.

5 · Evaluation

Why this matters: women rarely volunteer sexual problems, so ask directly and examine.

The **clinician must ask**, gender-neutrally, without assuming penile–vaginal intercourse.

> **ALGORITHM**
> 1. **P**ermission.
> 2. **L**imited **I**nformation: cycle, ageing, drugs.
> 3. **S**pecific **S**uggestions: lubricants, **vaginal oestrogen**, positions, vibrators.
> 4. **I**ntensive **T**herapy means **referral** (PLISSIT).

- **BSSC-W** (Brief Sexual Symptom Checklist for Women) is an ACOG triage tool, **not validated**. **DSDS** (Decreased Sexual Desire Screener) is **validated**, 5 items; yes to 1–4 with no contributing cause means **generalised acquired HSDD**.
- **FSFI** (Female Sexual Function Index): 19 items, 6 domains, most used. FSDS-R measures **distress**. **PISQ-12/IR** is the **only validated** tool for prolapse and incontinence. For HSDD: PFSF postmenopause, SIDI premenopause.
- History: lifelong or acquired, generalised or situational, and the **order problems appeared** (primary vs secondary); drugs, surgery, LUTS, prolapse, **partner's** function.
- Examine (essential with pain). Atrophy shows pallor, lost rugae, narrow introitus; **pH ≥4.5** without infection or semen means atrophy. Do a **Q-tip test** of the vestibule; **retract the hood** for adhesions.
- Lichen sclerosus (white plaques) leads to introital stenosis; lichen planus (**Wickham striae** pathognomonic) to vaginal stenosis.
- **Overactive** PFM causes dyspareunia; **underactive** PFM causes incontinence and prolapse. Fornix tenderness suggests endometriosis; ischial spine tenderness, pudendal neuropathy. Lukban–Whitmore tenderness runs 0 to 4 (prevents contraction). Do a cough stress test.
- Labs are rarely needed: HbA1c and lipids if at risk; prolactin/thyroid if signs. **No random testosterone**: no female range, unreliable assays, and no level predicts HSDD. Photoplethysmography/Doppler are research only.

6 · Hypoactive sexual desire disorder

Why this matters: HSDD is commonest and has licensed drugs, used only after reversible causes are treated.

- ISSWSH: ≥6 months lost motivation or desire, **with distress**, not due to drug, illness, pain, abuse or partner. **Exclude relationship conflict first.** Coexisting arousal or orgasm disorder does not exclude HSDD.
- Desire and distress both fall with age, so HSDD peaks in **midlife**. It is defined by distress, not hormone level; many have normal oestradiol.

ALGORITHM

1. Screen (DSDS), then history and examination.
2. **Correct modifiable factors**: drugs, depression, GSM, pain, partner's ED.
3. Education; sex or couples therapy.
4. If **generalised acquired** HSDD persists, then drug; review at 8 weeks.
5. **Refer** if situational HSDD fails counselling, or psychiatric illness or sexual trauma.

	Flibanserin	Bremelanotide
Mechanism	Central **5-HT1A agonist / 5-HT2A antagonist**: lowers serotonin, raises dopamine/noradrenaline (removes the brake)	Central **melanocortin agonist** releasing dopamine (presses the accelerator)
Licence	US Food and Drug Administration (FDA): only **premenopausal** acquired generalised HSDD	
Dose	**100 mg at bedtime** daily; ~half respond by 8 weeks	**1.75 mg subcutaneous** ≥45 min before sex; ≤1/24 h, **≤8/month**
Side effects	Dizziness, somnolence, nausea	Nausea, flushing, headache, **hyperpigmentation**

	Flibanserin	Bremelanotide
Cautions	**Alcohol** (hypotension, syncope); REMS restricted prescribing; †strong CYP3A4 inhibitors	Uncontrolled hypertension/CVD

- **Transdermal testosterone** (off-label) improves desire, arousal, orgasm and satisfaction in **postmenopausal HSDD**; oestrogen with androgen beats oestrogen alone. There is no cardiovascular signal; long-term safety is unknown.
- †2019 Global Consensus: target **premenopausal physiological levels**; check levels at baseline and 3–6 weeks; **stop at 6 months** if no benefit; watch for virilisation. Bupropion (off-label) is an alternative.

7 · Arousal disorder and PGAD

Why this matters: felt arousal and genital blood flow differ, so vasodilators rarely help.

- **Subjective and genital arousal correlate poorly**: vasodilators (L-arginine with yohimbine) raise flow without felt arousal. Failure with one partner only is a relationship problem.
- Use mindfulness-based sex-focused cognitive behavioural therapy (CBT); depression and anxiety predict poor response.
- **Transdermal testosterone** has the best evidence. PDE5 inhibitors help only with neurovascular impairment (type 1 DM, SCI, MS, SSRI-induced). **Bupropion 150 mg twice daily** treats SSRI-induced dysfunction. Topical alprostadil is ineffective; apomorphine has small studies.

PGAD (Leiblum–Nathan): genital arousal for hours to days, **not relieved by orgasm**, without desire, and distressing. Causes include Tarlov cysts, pelvic varices and **stopping an SSRI**. Treat with CBT, duloxetine or pregabalin; operate for a lumbosacral annular tear.

8 · Female orgasmic disorder

Why this matters: inadequate stimulation is common, so exclude it before diagnosing FOD.

- FOD is a **diagnosis of exclusion requiring distress**: exclude drugs, disease, stress and **inadequate stimulation** (including partner's ED/PE).
- **No orgasm at coitus without clitoral stimulation is normal variation.**
- Lifelong FOD reflects inexperience or poor education; acquired FOD reflects stimulation, context or physiology. Risks: anxiety, depression, sex guilt, conflict, abuse, DM, SCI, MS, cancer, SSRIs, antipsychotics, antiandrogens.
- **Education is the cornerstone**, with directed masturbation, sensate focus and mindfulness. If arousal is intact, increase stimulation (vibrators; MCQ EROS clitoral vacuum is the only FDA-cleared FSD device). Give hormones for postmenopausal deficiency; operate for clitoral phimosis or adhesions.

ALGORITHM

1. **SSRI-induced dysfunction**: was it present before treatment?
2. If not, wait for tolerance (~4 months).
3. If it persists at 3–6 months, then lower the dose, or switch to or add **bupropion** or mirtazapine; consider sildenafil.

9 · Genito-pelvic pain / penetration disorder

Why this matters: pain usually responds to pelvic floor therapy rather than surgery.

- **Dyspareunia** is superficial or deep (endometriosis, prolapse, pelvic floor). **Vaginismus** is involuntary contraction preventing penetration.
- **Vulvodynia**: vulval (burning) pain for **≥3 months**, no cause; localised or generalised, provoked or unprovoked.
- **Provoked vestibulodynia (PVD)** is commonest (~**10–12%** of women; linked to irritable bowel syndrome, fibromyalgia). Primary PVD starts at first penetration (nulliparity, family history, enuresis, dysmenorrhoea). Secondary (~⅔) follows pain-free sex; its area is larger, and the hood is often most painful.
- Mechanism: vestibular nerve proliferation, with a **hypertonic pelvic floor** and fear–avoidance.
- **Clitorodynia**: hood adhesions trap keratin, forming painful pseudocysts and pearls, so **examine the glans routinely**.

ALGORITHM

1. Education and vulval care.
2. Lidocaine 5% or EMLA **10 min before intercourse**.
3. **Multimodal pelvic floor physical therapy** (myofascial release, trigger points, down-training, biofeedback, graded **dilators**, electrical stimulation) with psychological/couples therapy.
4. If hypo-oestrogenic, then vestibular oestrogen.
5. If localised PVD is refractory, then **vestibulectomy**.

- **Multimodal physical therapy is standard of care** for PVD; EMG biofeedback is best if the pelvic floor is overactive.
- Oestrogen deficiency **increases** vulval nerve density, so **topical beats systemic and vestibular beats vaginal**.
- †Second line: amitriptyline, gabapentin, intralevator botulinum toxin.
- **Vestibulectomy** extends to within 1–2 mm of the meatus; it beats non-surgical therapy in localised neuroproliferative disease.

Treatment of GSM

ALGORITHM

1. Non-hormonal lubricants/moisturisers and regular activity (NAMS first line).
2. Then **low-dose vaginal oestrogen** (†minimal absorption, no progestogen) or †vaginal DHEA (prasterone 6.5 mg nightly).
3. If vaginal products are impossible, then †ospemifene 60 mg orally (SERM, vaginal agonist; VTE risk).
4. If vasomotor symptoms, then systemic hormone therapy.

KEY POINT

†After **breast cancer**, use non-hormonal measures first; vaginal oestrogen only after oncology discussion (aromatase inhibitors).

10 · Pelvic surgery and sexual function

Why this matters: pelvic operations can damage nerves, blood supply or vaginal shape; technique and counselling limit harm.

Surgery harms by three routes: **nerves** (lateral vaginal wall, pelvic plexus), **clitoral blood supply** (distal urethra) and **vaginal shape**. Counsel **before and after**; patients want it more than they ask.

- **Radical cystectomy**: loss of desire is commonest, then orgasm, dyspareunia and lubrication problems; ~half can have intercourse. Preserve function by (1) retrograde dissection saving the bundles on the **lateral vaginal walls**, (2) dividing the urethra at the **bladder neck** (clitoral supply), and (3) keeping vaginal dimensions.
- **Genital-sparing** cystectomy (if oncologically safe) lets sex resume at **~6 vs 14 weeks**; diversion type is irrelevant. A third of cancer patients want sex-therapy referral. **Never trade oncological control.** *See Ch 101*
- **Hysterectomy**: benign surgery leaves function unchanged or better, whatever the route. After radical surgery, lubrication problems and dyspareunia dominate at 6 months; most are active by 12 months.
- **Rectal surgery**: dysfunction is common after low anterior resection and occurs in **nearly all** after abdominoperineal resection. Risks: neoadjuvant chemotherapy, low tumour, intersphincteric resection, Hartmann's.
- **Prolapse (POP) repair**: stage does not predict function; bother and body image do. After native-tissue repair most are same or better; de novo dyspareunia is possible.
- Dyspareunia is commoner after vaginal than abdominal repair, and after anterior mesh than **sacrocolpopexy** (apical gold standard). Biological grafts recurred more and were abandoned. Mesh complications may appear years later.
- **Combined anterior and posterior repair increases dyspareunia**, so stage it in sexually active women. **Keep length, avoid excess skin excision, preserve the introitus.** *See Ch 119*
- **Stress urinary incontinence (SUI) surgery**: ~⅓ improve, most are unchanged; **curing coital incontinence** predicts improvement.
- Retropubic tape (**TVT**) may denervate the vaginal wall; transobturator tape (**TOT**) spares the dorsal clitoral nerve. PISQ-12 slightly favours TOT; FSFI shows no difference. Dyspareunia is a strong reason for sling removal.
- **Coital incontinence** (~1 in 10 incontinent women): leakage at penetration is stress; at orgasm, detrusor overactivity (DO).
- FSD severity by lower urinary tract dysfunction: bladder pain syndrome (BPS) > DO > urgency/mixed UI > SUI > dry overactive bladder (OAB) > voiding LUTS. HSDD pairs with SUI, orgasm problems with OAB.
- **BPS is worst**: pain causes fear, then PFM overactivity, then vulvodynia. Tolterodine improved all FSFI domains in OAB; **sacral neuromodulation** improves sexual function whatever the cause.
- **Genital cosmetic surgery** has poor evidence; screen for **body dysmorphic disorder**.

11 · Special populations

Why this matters: transgender and spinal-injured women have specific risks.

Ask orientation and gender identity; a **trans male** was assigned female at birth. Offer fertility preservation before transition. Oestrogen risks thromboembolism; testosterone risks erythrocytosis (haematocrit **>50%**), so check haematocrit and testosterone 3-monthly in year 1, then 1–2 times a year. Spironolactone needs potassium checks. SCI impairs sensation, lubrication and orgasm, with genital pain and spasticity. *See Ch 126*

12 · Landmark trials

Trial	Design	Result	Take-home
Flibanserin phase III	RCTs vs placebo, premenopausal	More satisfying events and desire, less distress	First FDA-approved drug for HSDD
RECONNECT (bremelanotide)	2 phase III RCTs	More desire, less distress	On-demand option for premenopausal HSDD
Bupropion for SSRI-induced dysfunction (Cochrane)	Meta-analysis	Benefit at 150 mg twice daily	Add or switch to bupropion

★ Putting it together: female pelvic decisions

A · Incontinent woman

Why this matters: the type of incontinence decides treatment.

ALGORITHM

1. **Classify** (history, diary): stress (SUI), urgency (UUI), mixed, overflow, functional. Check stress test, mobility, prolapse (POP), residual and urinalysis (*Ch 105*).
2. **Urodynamics** are not needed for uncomplicated demonstrable SUI (VALUE). If mixed or unclear, voiding dysfunction, prior surgery, neurological or high-grade POP, then use them.
3. **Conservative first:** weight, fluids/caffeine, supervised pelvic floor training, bladder training, pessary, postmenopausal vaginal oestrogen (*Ch 114*).
4. If this fails, then **SUI surgery** (*Ch 116, 118*); for UUI, drugs, botulinum toxin, neuromodulation.

B · Choosing an SUI operation

Why this matters: match the operation to the patient.

Operation	Best for	Drawbacks
Retropubic mid-urethral sling (MUS)	Standard; better for intrinsic sphincter deficiency	Perforation (cystoscopy mandatory), voiding dysfunction, rare vascular/bowel injury
Transobturator MUS	Avoiding retropubic space (prior surgery, obesity)	Groin/thigh pain; less effective in severe SUI
Autologous pubovaginal sling	Recurrent SUI, post-mesh, with urethral repair, mesh declined	Larger; more voiding dysfunction
Burch	Mesh declined; concurrent abdominal surgery	Posterior prolapse, de novo urgency
Bulking agent	Frail, minimal surgery	Lower, short-lived cure

C · Prolapse: which repair?

Why this matters: treat symptoms, and always support the apex.

ALGORITHM

1. **POP-Q** stages the leading edge; stage II is within 1 cm of the hymen (*Ch 105*).
2. If asymptomatic, then observe; if symptomatic, then pessary or surgery.
3. **Support the apex**: uterosacral/sacrospinous suspension or sacrocolpopexy (most durable) (*Ch 117*).
4. If **frail, not sexually active**, then colpocleisis.
5. If **occult SUI**, then discuss a concomitant sling (OPUS: fewer leaks, more complications).

D · Vesicovaginal fistula

Why this matters: the first repair has the best chance.

KEY POINT

Do a double-dye test, upper tract imaging, and biopsy after radiation or cancer. Post-hysterectomy fistulae are supratrigonal and mostly repaired vaginally (Latzko or layered); go abdominal if the ureter is involved, the fistula is large or high, or vaginal repair failed. Use tension-free watertight layers, non-overlapping lines, a vascularised flap (Martius vaginally, omentum abdominally) for poor or irradiated tissue, continuous drainage, then a cystogram (*See Ch 121*).

E · Mesh complications

Why this matters: problems may appear years later.

ALGORITHM

1. If suspected, then speculum and cystoscopy.
2. If the exposure is small, then oestrogen with or without trimming; if larger or symptomatic, then excise and close.
3. If mesh is in bladder or urethra, then remove and repair; no new mesh then.
4. Remove enough to solve the problem; never asymptomatic mesh (*Ch 119*).